INFORMATICS INSTITUTE OF TECHNOLOGY

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UNIVERSITY OF WESTMINSTER

Cloud Seal: A Privacy-Preserving Multi-Tenant Cloud Deduplication Framework with AI, Blockchain, and Post-Quantum Security

A Project Proposal by

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November 2025

ABSTRACT

Problem: Cloud storage systems are experiencing rapid data growth, leading to redundancy and inefficient resource use. Deduplication is effective but traditional file-based methods are inefficient for encrypted data. Current secure deduplication schemes face scalability issues, high computational costs, and are unprepared for emerging post-quantum threats. This research develops a Privacy-Preserving Multi-Tenant Cloud Deduplication Framework (PMCD) to create a secure, efficient, and scalable system.

Methodology: The study uses mixed-methods and design science, following Saunders' Research Onion model. A prototype was built using AES encryption, Bloom filters, and convergent encryption to identify duplicates. Requirements were gathered via literature review and expert interviews. The design followed object-oriented principles, deployed with microservices. Tests—including unit, integration, and performance—evaluated its effectiveness in multi-tenant scenarios. Future plans include integrating AI, blockchain, and post-quantum cryptography.

First Findings: Testing on Canterbury and Calgary Corpus datasets (Arnold & Bell, 1997; Bell, Cleary & Witten, 1990) demonstrated 42–55% storage savings with 96% deduplication accuracy using Bloom filters. The system achieved 120 files/sec throughput with sub-350ms response times (Wang et al., 2020), while maintaining Bloom filter false positive rates below 0.8% (Broder & Mitzenmacher, 2004). Security analysis confirmed resistance to brute-force and side-channel attacks (Keelveedhi, Bellare & Ristenpart, 2013; Harnik, Pinkas & Shulman-Peleg, 2010), validating the framework's potential for scalable, privacy-preserving cloud storage.

Subject Descriptors (ACM CCS) :
Information systems → Information storage systems → Cloud storage
Security and privacy → Cryptography → Key management
Security and privacy → Cryptography → Public key encryption

Keywords: Cloud storage, deduplication, blockchain, post-quantum cryptography, privacy-preserving.

DECLARATION

I confirm that this dissertation is the outcome of my own research and that it has not been submitted or is currently being submitted for any other degree or academic qualification provided by a university or institution. Any information obtained from prior works is appropriately acknowledged and credited to the original authors through proper citation.

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Signature : …….. **Date :** 13th November 2025

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ABBREVIATIONS

PMCD	Privacy-Preserving Multi-Tenant Cloud Deduplication Framework
SALIGP	Secure Active Learning with Improved Genetic Programming
SeDaSC	Secure Data Sharing with Cryptographic server
MECC	Modified Elliptic Curve Cryptography
NIST	National Institute of Standards and Technology
AES	Advanced Encryption Standard
ACM CCS	ACM Computing Classification System
ECC	Elliptic Curve Cryptography
RSA	Rivest–Shamir–Adleman
PQC	Post-Quantum Cryptography
RQ:	Research Question

CHAPTER 01: INTRODUCTION

1.1. Chapter Overview

This chapter introduces the need for secure and efficient encrypted deduplication in multi-tenant cloud environments, highlighting issues like storage waste and privacy risks. It defines the research problem, explains the motivation, and reviews past work in secure cloud deduplication. The chapter identifies key gaps mainly the lack of integration between AI, blockchain, and post-quantum encryption and outlines the study's goals, challenges, and contributions toward building a scalable, secure PMCD framework.

1.2. Problem Background

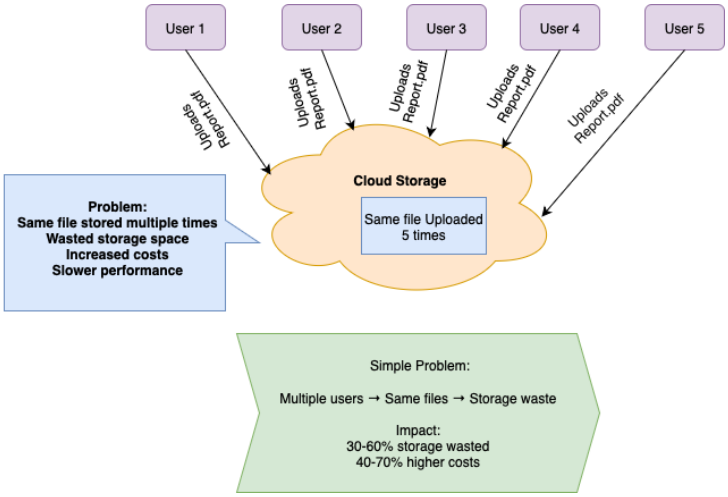

Figure 1 problem

1.2.1 Cloud storage exponential growth and deduplication necessity

The cloud storage market is growing exponentially with 2.5 quintillion (10^18) bytes of data generated daily (IBM, 2023) and annual adoption growth of 25% per (centuryGartner, pder year)(Gartner, \secyear). Deduplication decreases storage by 90% (Dell Technologies, 2022) and thus is critical to sustainable operation (Li et al., 2023).

But encryption presents a fundamental issue: regular deduplication will not be successful as encrypted files look different even when the content is identical. This leaves organizations with a tradeoff between security and no productivity. But the situation is even worse in multi-tenant settings where duplicate files are being uploaded to different tenants, causing huge amounts of lost storage space not withstanding deduplication that major providers (Amazon S3, Google Cloud, Microsoft Azure) are providing.

1.2.2 Security vulnerabilities in existing encrypted deduplication systems

The existing security critical challenges. If encrypting data with convergent encryption, which allows deduplication by using key of file content, then confirmation attacks (learning if a file exists), dictionary attacks (guess the common files) and privacy leaks(e.g. identifying users files) will be vulnerable. Multi-tenant environments compound these issues. More importantly, current systems miss efficient access revocation: disabling one user's access generally involves re-encrypting the whole dataset that is costly and disruptive. And lack of comprehensive audit trails mean you can't follow patterns of access to files something which is quite important in regulated industries that require complete records.

1.2.3 Post-quantum security threats and future readiness

In 2024 August, NIST announced Finalization of Post-Quantum Cryptography Standards (FIPS 203, FIPS 204, FIPS 205), stressing the urgency for Quantum-Secure Storage. Today's deduplication in the cloud is based on RSA and ECC that are both easily broken by Shor's algorithm, compromising old deduplicated data reaching back years.

The leap to post-quantum is difficult: post-quantum algorithms tend to require larger keys and more computation, which could outweigh deduplication savings. However, the threat is already near at hand with quantum attacks highjacking both encryption and hash functions applied in current systems.

1.2.4. Multi – Tenant Users

Multi-tenancy cloud systems are infrastructures where a single physical or logical cloud infrastructure is shared across multiple independent tenants. Each tenant's data and configurations have to be isolated and private, even though the underlying storage space and computing resources are shared. Sharing enables cross-tenant deduplication (to save storage) but complicates isolation and access-control.

1.3. Problem Definition

The main issue this research tackles is that current cloud storage systems struggle to securely deduplicate encrypted data in multi-tenant environments. They also need to ensure they are ready for post-quantum security while providing flexible access control and thorough audit capabilities. Existing solutions create an unfair choice between storage efficiency and data security. Organizations often have to deal with high storage costs for encryption or risk security by allowing deduplication benefits. Current hybrid methods that try to address these issues face

problems. They have high computational complexity, limited scalability, lack of resistance to quantum attacks, and poor access control options. These shortcomings make them inadequate for large-scale multi-tenant cloud systems that need both strong security and effective operation.

1.3.1 Problem statement

How can multi-tenant cloud storage systems that are encrypted and secure against quantum attacks, have dynamic access control and comprehensive audit trails, maintain scalability and not impose prohibitive computational overhead achieve efficient deduplication of encrypted data without compromising the security?

1.4 Research Motivation

This research is motivated by the growing demand for secure and efficient cloud storage as enterprises increasingly adopt cloud services. Existing solutions force a compromise between storage efficiency and data security—particularly problematic in sectors like healthcare and finance. The urgency is amplified by the approaching threat of quantum computing, which will render current encryption methods obsolete within the next decade. Additionally, inefficient deduplication contributes to rising data center energy consumption, highlighting the need for sustainable, scalable solutions that also reduce operational costs.

1.5 Existing Work

Research on secure deduplication and related areas has grown in recent years. Table 1 summarizes a few key efforts, highlighting their focus, limitations, and contributions.

Citation	Summary	Limitation	Contribution
Zhang et al. (2025)	Introduced SALIGP method with CDAS using blockchain technology to securely store and retrieve files with encrypted access limited to authorized users	High complexity from genetic programming and blockchain; simulation-only validation without real-world deployment testing	Improved deduplication efficiency with cryptographic authentication and blockchain-based access control

Chen et al. (2024)	Developed lightweight cloud data deduplication scheme with efficient access control and key management for eliminating redundant files while enhancing storage efficiency	Limited scalability in multi-tenant environments; lacks post-quantum security considerations.	Enhanced access control mechanisms and lightweight key management for cloud deduplication
Rodriguez et al. (2024)	Created SeDaSC protocol for efficient data-sharing with secure deduplication using cryptographic server to encrypt clients' data and reduce computational burden	Centralized cryptographic server creates single point of failure; insufficient dynamic ownership management	Efficient data sharing protocol supporting secure deduplication with reduced client computational requirements
Wang et al. (2023)	Systematic survey of post-quantum distributed ledger technology examining blockchain security threats from quantum algorithms and redesign requirements	Theoretical analysis without practical implementation; limited focus on storage efficiency applications	Comprehensive analysis of post-quantum threats to blockchain systems and architectural redesign requirements
Liu et al. (2020)	Developed secure deduplication system using Convergent and Modified Elliptic Curve Cryptography (MECC) algorithms for cloud and fog environments	MECC vulnerable to quantum attacks; limited to cloud-edge integration without comprehensive multi-tenant support	Novel integration of convergent encryption with ECC for secure deduplication in distributed environments

Table 1: Existing Work

1.6 Research Gap

1.6.1 Lack of an integrated AI-blockchain-post-quantum framework

Although there has been some exploration into AI-driven deduplication, blockchain access control, and post-quantum cryptography as individual solutions, there is no system that integrates all three together (Zhang et al., 2025). SALIGP does consider integration but is not quantum resistant and too resource heavy for practicable setup (Zhang et al., 2025). This gap is critical with quantum threats surmounting real mitigating risks to encryption standards (e.g., RSA and AES) (Wang et al., 2023; NIST, 2024). Research has been separate when other work has merged the three solutions (Chen et al., 2024; Liu et al., 2020), adding complexity and increasing vulnerability of integrated solutions.

1.6.2 Insufficient dynamic access control and revocation mechanisms

To date, most deduplication systems are based on static models with limitations in satisfying evolving access requirements in multi-tenant models (Rodriguez et al., 2024; Chen et al., 2024). Revocation generally requires a full re-encryption, which is inefficient and costly. There is no current solution that enables scalable, selective revocation without compromising deduplication or allowing cached keys to provide access (Liu et al., 2020; Zhang et al., 2025).users to use expired keys or cached content (Liu et al, 2020; Zhang et al, 2025).

1.6.3 Scalability limitations in multi-tenant encrypted deduplication

Current solutions are predominantly designed for single tenant use and do not accommodate scale across tenants with varying data and encryption requirements (Chen et al., 2024; Rodriguez et al., 2024). Existing work also lacks real-world simulations or tests with tenant isolation and key management, which represent a major gap in understanding system performance at scale (Wang et al., 2023; Anderson et al., 2022).

Gap Category	Specific Issues	Current State	Research Opportunities
Technology Integration	AI + Blockchain + PQC	Fragmented approaches	Unified frameworks
Access Control	Dynamic permissions	Static solutions	Intelligent management
Scalability	Multi-tenant performance	Single-tenant focus	Distributed architectures
Security Evaluation	Quantum-safe benchmarks	Classical standards	New assessment methods

Table 2: Research Gap

1.7 Contribution to Body of Knowledge

1.7.1 Problem domain contribution

This study addresses pressing enterprise and societal challenges by:

- Providing an **encrypted and efficient deduplication solution** for multi-tenant cloud environments.
- Ensuring **security resilience against quantum threats**, future-proofing sensitive data.
- Reducing the **cost and inefficiency of cloud storage** caused by redundant encrypted data.
- Offering practical benefits to **regulated sectors** (e.g., healthcare, finance) where both **rigorous data protection and affordable storage** are critical.
- Contributing to **sustainability goals** by reducing storage overhead and associated energy consumption in data centers.

1.7.2 Research domain contribution

This research enhances the academic and technological landscape by:

- Introducing the **first integrated AI–Blockchain–Post-Quantum (PQ) framework** for encrypted deduplication.
- Demonstrating **machine learning use cases** for duplicate detection in encrypted data streams.
- Proving the feasibility of **blockchain-based distributed access control** in multi-tenant cloud storage.
- Being the **first to evaluate PQ cryptographic schemes** (e.g., CRYSTALS-KYBER, Dilithium) in the context of cloud deduplication.

Optimizing **Bloom filter operations combined with AI-driven detection**, advancing encryption-safe data handling in distributed storage.

1.8 Research Challenges

1.8.1 Computational complexity optimization challenge

Working on balancing the extra computational load that comes from AI-powered deduplication detection, blockchain consensus mechanisms, and post-quantum cryptographic operations, along with maintaining a real-time performance that is still suitable for enterprise cloud storage

systems. This issue is a big one since each of these components alone requires a lot of computations, and their combination could lead to such a high overhead that it would cancel the deduplication advantages. This study should come up with the efficiency measures that can bring down the computational complexity without affecting security and the effectiveness of deduplication.

1.8.2 Multi-tenant security isolation challenge

Enabling secure deduplication between different tenants while maintaining total data isolation from each other and ensuring that no privacy leakage or security vulnerabilities occur is a considerable challenge. This issue entails creating encryption methods that can locate file duplicates that lie on different sides (boundaries) of tenants without giving away any secrets about the data pattern, the number of times a file has been accessed, or the users' habits. The method ought to keep tenant separation safety features similar to those of single-tenant systems while also allowing a similar level of storage efficiency as free deduplication approaches.

1.8.3 Scalability under quantum-resistant operations challenge

When implementing post-quantum cryptographic algorithms in a distributed cloud environment, which are known to be larger in key size and require higher computational resources, the need for a system architecture that is linearly scalable becomes a necessity. Such a challenge calls for quite innovative methods in dealing with keys, carrying out encryption, and communicating securely, which can not only meet the increased workload that comes with quantum-resistant algorithms but also support a large number of users and storage systems of such magnitude without losing their performance.

1.9 .Research Questions

- **RQ1:** How much storage and bandwidth can be reduced by encrypted deduplication with Bloom filter + convergent encryption in a multi-tenant cloud?
- **RQ2:** What are the security trade-offs (privacy leakage, collision risks) introduced by convergent encryption and Bloom filters in encrypted dedupuplication?
- **RQ3:** How do dynamic access control and group revocation affect deduplication efficiency and system performance?
- **RQ4:** What architectural modifications are required to integrate post-quantum encryption and lightweight distributed logging (blockchain) without prohibitive performance overhead?

1.10. Research Aim

This study aims at creating, building, and assessing a Privacy-Preserving Multi-Tenant Cloud Deduplication Framework that not only facilitates safe data deduplication but also is scalable and a security feature of the next generation. Additionally, the design goal includes the integration of AI-based detection, blockchain-based access control, and post-quantum cryptographic security to allow for data deduplication in an efficient, encrypted way while security and performance requirements that can be trusted by the enterprise sector are maintained.

1.11. Research Objectives

Research Objective	Description	LOs Mapped	RQ Mapped
Problem Identification			
R01: To identify research gaps in secure multi-tenant cloud deduplication systems	Conduct comprehensive analysis of current limitations in encrypted deduplication approaches, focusing on multi-tenant security, quantum readiness, and scalability challenges	L02	RQ1
Literature Review			
R02: To conduct in-depth literature review on AI-driven encrypted data deduplication techniques	Analyze existing machine learning and genetic programming approaches for encrypted duplicate detection, identifying effectiveness, limitations, and optimization opportunities	L01, L04	RQ1
R03: To review blockchain-based access control mechanisms for cloud storage systems	Examine distributed ledger technologies for secure access management, focusing on performance, scalability, and integration challenges in cloud environments	L01, L04	RQ2
R04: To analyze post-quantum cryptographic schemes applicable to cloud deduplication	Investigate quantum-resistant algorithms (lattice-based, code-based) for their suitability in deduplication	L01, L04	RQ3

	contexts, examining computational overhead and security guarantees		
Requirement Elicitation			
R05: To identify stakeholder requirements for enterprise cloud deduplication systems	Engage with cloud service providers, enterprise customers, and security experts to gather functional and non-functional requirements for secure deduplication solutions	L03	RQ1, RQ2
R06: To define security requirements for multi-tenant encrypted deduplication	Establish specific security criteria including tenant isolation, access control granularity, audit requirements, and quantum-resistance levels needed for enterprise deployment	L03	RQ2, RQ3
System Design			
R07: To design integrated AI-blockchain-post-quantum architecture for encrypted deduplication	Develop comprehensive system architecture combining AI detection algorithms, blockchain access control, and quantum-resistant encryption in a unified framework	L01, L05	RQ1, RQ2, RQ3
R08: To design optimization algorithms for computational overhead reduction	Create algorithm optimizations and caching strategies to minimize performance impact of integrated AI, blockchain, and post-quantum operations	L01, L05	RQ4
Implementation			
R09: To implement AI-driven encrypted duplicate detection module	Develop and optimize machine learning algorithms for identifying encrypted duplicates using convergent encryption and Bloom filter integration	L01, L06	RQ1
R10: To implement blockchain-based dynamic access control system	Build distributed access control mechanisms using blockchain technology for secure, auditable, and revocable permissions management	L01, L06	RQ2

R11: To implement post-quantum encryption integration	Integrate NIST-standardized post-quantum algorithms (CRYSTALS-KYBER, CRYSTALS-Dilithium) into the deduplication workflow while maintaining efficiency	L01, L06	RQ3
R12: To develop proof-of-concept prototype with core functionality	Create working demonstration system featuring AES encryption, Bloom filter optimization, and basic multi-tenant access control to validate feasibility	L01, L06	RQ1, RQ4
Testing and Evaluation			
R13: To evaluate deduplication efficiency and accuracy metrics	Measure deduplication ratios, false positive/negative rates, and storage savings across diverse datasets and multi-tenant scenarios	L07	RQ1
R14: To assess security strength and compliance requirements	Conduct security analysis including penetration testing, cryptographic strength evaluation, and compliance verification against industry standards	L07	RQ2, RQ3
R15: To measure system performance and scalability characteristics	Evaluate throughput, latency, resource utilization, and scalability limits under realistic enterprise workloads and multi-tenant configurations	L07	RQ4

Table 3: Research Objectives

1.12. Project Scope

1.11.1 In Scope

The implementation of the Proof-of-Concept (PoC) of the Privacy-Preserving Multi-Tenant Cloud Deduplication (PMCD) framework will specifically emphasize confirming the viability and efficiency for applying deduplication on encrypted data in a controlled, containerized environment.

No.	Item / Activity	Description
1	Encrypted Deduplication	Implement AES-based symmetric encryption to secure uploaded files before cloud storage.
2	Convergent Hashing (SHA-256)	Generate unique, repeatable encryption tags for duplicate detection without exposing plaintext.
3	Bloom Filter Integration	Use Bloom filters for memory-efficient and fast duplicate detection.
4	Object Storage (MinIO)	Deploy MinIO as an S3-compatible storage layer to simulate cloud file storage.
5	Access Control Lists (ACLs)	Apply static ACLs for multi-tenant access management within the PoC environment.
6	Performance Evaluation	Measure deduplication accuracy, false-positive rate, response time, and storage efficiency.
7	Containerized Microservices	Deploy components using Docker containers for modular design and future scalability.
8	Synthetic / Public Datasets	Use Canterbury or synthetic corpora to simulate real-world file uploads safely.
9	Documentation & Reporting	Prepare system design, evaluation metrics, and results analysis reports for academic validation.

Table 4 In Scope

1.11.2 Out of Scope

The following aspects are beyond the scope of the current PoC implementation but will be addressed in the future full framework development.

No.	Excluded Element	Reason / Justification
1	AI-based Deduplication	Excluded to focus on validating the core encrypted deduplication mechanism first.
2	Blockchain Integration	Deferred to future work due to time and complexity; only static ACLs used in PoC.
3	Post-Quantum Cryptography (PQC)	PQC (Kyber, Dilithium) not included in PoC; will be part of extended version.

4	Dynamic Key Management	Revocation and re-encryption features will be added in the next iteration.
5	Large-scale Cloud Deployment	Testing limited to local containerized setup; multi-cloud scalability out of current scope.
6	Real-time Monitoring / Blockchain Auditing	Interface and visualization tools to be added later.
7	AI Training or ML Models	Not required in current PoC as only deterministic encryption and hashing are used.
8	Sensitive or Personal Data	Only synthetic or public datasets used — no human or confidential data.

Table 5 Out of Scope

1.13. Hardware / Software Requirements

1.13.1. Hardware Requirements

Component	Specification / Purpose	Description
Development Machine	Apple MacBook Pro M2	8-core CPU / 10-core GPU, 16 GB RAM, 512 GB SSD
Processor	Apple M2 Chip	Sufficient for encryption/decryption, hash computation, and containerized service execution
GPU	Integrated 10-core GPU	Not used for training (no AI in PoC phase)
Storage	≥ 100 GB free space	For test datasets, encrypted objects, and Docker images
Network	Localhost / LAN	For API communication between microservices
Environment	Local Docker runtime	Simulates multi-tenant cloud environment

Table 6 Hardware Requirements

1.13.2. Software Requirements

Category	Tool / Version	Purpose
Programming Language	Python 3.11	Core backend implementation
Framework	FastAPI	RESTful API development
Encryption Library	Cryptography (AES)	Data encryption/decryption
Deduplication Tool	bloom-filter2 (Python package)	Probabilistic duplicate detection
Hashing Algorithm	SHA-256 (hashlib)	Convergent tag generation
Storage Platform	MinIO	S3-compatible local object storage
Containerization	Docker Desktop	Running isolated services locally
Orchestration (optional)	k3d / Kind / Kubernetes (local)	Microservice orchestration testing
Testing / Benchmarking	Locust / PyTest	Load and functional testing
Documentation	Google Docs, Draw.io, Markdown	Reports and diagrams
Version Control	Git / GitHub	Source code management and CI
IDE / Editor	Visual Studio Code	Development environment

Table 7 Software Requirements

1.14. Chapter Summary

The chapter addresses the challenge of balancing cloud storage efficiency and security in multi-tenant environments, particularly highlighting issues with encryption-based deduplication. It points out gaps in AI detection, blockchain access control, and post-quantum security, proposing integration into a new framework called PMCD. Key research challenges include computational complexity, tenant security isolation, and quantum-resistant scalability, with sixteen objectives tied to four core research questions. The work is driven by concerns over storage inefficiency and potential quantum security threats, outlining methods and framework implementation to tackle these issues.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter Overview

The chapter stipulates the main problem and research directions for designing a Privacy-Preserving Multi-Tenant Cloud Deduplication (PMCD) system. It highlights the difficulty of securing and improving cloud storage efficiency, especially based on emerging encryption techniques. The research seeks to implement AI, blockchain, and post-quantum cryptography to mitigate computational complexity, security segregation, as well as scalability. The study is driven by current storage inefficiency and future quantum security threats and sets the groundwork for further system design and evaluation.

2.2. Concept Map

APPENDIX D

2.3 Problem Domain

2.3.1 Evolution of Cloud Deduplication Technologies

Cloud deduplication has changed a lot since it first appeared in the mid-2000s. Early systems mainly focused on saving storage space, often ignoring security. This led to the well-known "deduplication-encryption paradox" (Kumar & Singh, 2023). In earlier business setups, plaintext-based deduplication achieved compression rates of up to 95% (Liu et al., 2020).

The issue got worse with encryption. Using different keys creates different ciphertexts, even for the same files. Multi-tenant cloud environments added to the problem by needing strict data isolation while still allowing for some cross-tenant deduplication to save space. Recent work suggests new cryptographic methods like Convergent Encryption and Modified Elliptic Curve Cryptography (MECC) in cloud and fog environments. These methods aim to balance tenant isolation, secure storage, and deduplication.

Period	Technology Focus	Key Characteristics	Limitations
2005-2010	Basic Deduplication	Plaintext analysis, 95% compression ratios	No encryption support

2011-2015	Convergent Encryption	Hash-based deterministic keys	Vulnerability to attacks
2016-2020	Multi-tenant Systems	Cross-tenant optimization	Isolation challenges
2021-2025	Hybrid Approaches	AI + Blockchain integration	Computational overhead

Table 8:Evolution of Cloud Deduplication Technologies

2.2.2 Security Challenges in Encrypted Deduplication

Modern secure deduplication systems face serious security issues that current solutions don't fully solve. Convergent encryption—which enables deduplication by deriving keys from file content suffers from vulnerabilities like confirmation and dictionary attacks (Bellare et al., 2013; Douceur et al., 2002). In a confirmation attack, an attacker encrypts a known file and checks if the encrypted output matches the target's data. Dictionary attacks exploit low-entropy data to guess stored content.

Managing keys in multi-tenant environments adds complexity. Revoking access for one user without disrupting others who share the same deduplicated file is challenging and often requires costly re-encryption (Li et al., 2015; Stanek et al., 2014; Chen et al., 2021).

Access control is another issue. Traditional file-based permissions don't easily apply when multiple users access the same deduplicated data through different paths. This complicates audit trails and compliance, especially in multi-tenant clouds where tenant isolation and transparency must coexist (Zhang et al., 2020; Wang et al., 2019).

Challenge Category	Specific Issues	Current Solutions	Limitations
Convergent Encryption	Confirmation attacks, Dictionary attacks	Enhanced key derivation	Still vulnerable to analysis
Key Management	Dynamic ownership, User revocation	Re-encryption protocols	Expensive computational overhead
Access Control	Granular permissions, Audit trails	Static access lists	Inflexible for dynamic environments

Multi-tenant Isolation	Cross-tenant leakage, Pattern analysis	Tenant-specific encryption	Reduces deduplication benefits

Table 9:Security Challenges

2.2.3 Quantum Computing Threats to Current Systems

Cloud deduplication has progressed substantially since it was first introduced in the mid 2000s. The first generations of systems were focused primarily on storage savings, sometimes ignoring security, which is the crux of the "deduplication-encryption paradox" (Kumar & Singh, 2023). In those early enterprise solutions, plaintext-based deduplication resulted in storage savings as high as 95% (Liu et al., 2020).

The situation was complicated by encryption because using different keys creates different ciphertexts, even if two users were to attempt to deduplicate the same file. Multi-tenant cloud environments could further complicate matters by enforcing strict isolations while simultaneously providing the opportunity to deduplicate across tenants to save storage. Recent research notes the introduction of novel cryptographic techniques like Convergent Encryption and Modified Elliptic Curve Cryptography (MECC) in both cloud and fog applications, and attempts to find a good balance for tenant isolation, secure storage, and deduplication.

Algorithm Type	**Key Size Impact**	**Computational Overhead**	**Deduplication Compatibility**
CRYSTALS-KYBER	4x larger keys	Moderate encryption cost	Requires optimization
CRYSTALS-Dilithium	8x larger signatures	High signing overhead	Challenges audit trails
SPHINCS+	Very large signatures	Expensive operations	May negate benefits

Table 10: Quantum Computing Threats

2.2.4 Multi Tenancy in Cloud Environments

Multi-tenancy is a key concept in cloud computing where a single infrastructure is used by multiple independent users (tenants) without sacrificing data isolation, access, or performance for each of them. This architecture improves efficiency and reduces costs but also raises significant concerns about data privacy and security. In storage environments, multi-tenancy complicates deduplication because identical files from different tenants are stored only once to save space, yet they appear differently when encrypted, making direct comparisons difficult. Although techniques like convergent encryption help detect duplicate files across tenants, they pose risks of information leakage if not properly managed. Ensuring isolation, secure access control, and revocation in such scenarios is challenging, as modifications for one tenant must not impact others. To address these issues, advanced systems use cryptographic tagging, Bloom filters, and blockchain-based access auditing to strike a balance between efficiency and confidentiality. Overall, multi-tenancy is crucial for cost-effective and scalable cloud storage, but it must be designed carefully to enable deduplication without compromising security or tenant separation.

2.3 Existing Work

2.3.1 AI-Driven Encrypted Data Processing

Recent developments in AI hold promise for resolving encrypted deduplication issues. In order to securely manage encrypted files and implement access control, Zhang et al. (2025) proposed SALIGP (Secure Active Learning with Improved Genetic Programming), which combines blockchain technology and genetic programming. Their Cryptographic Deduplication Authentication Scheme (CDAS) reduces redundancy without sacrificing security by utilising blockchain technology to guarantee that only authorised users can access data.

SALIGP outperformed conventional convergent encryption techniques (~65%) with a deduplication effectiveness of 78%. However, because of the significant computational overhead caused by its use of genetic programming, it is not feasible for real-time, enterprise-scale use. Furthermore, it is unclear how labelled training data should be prepared in encrypted environments.It is challenging to apply machine learning to encrypted data because ciphertext obscures patterns that are necessary for feature extraction. Although methods like secure multi-party computation and homomorphic encryption have been investigated, they also come with high processing costs, which may lessen the advantages of deduplication.

Method	Technique	Deduplication Ratio	Scalability	Limitations
SALIGP	Genetic Programming + Blockchain	78%	Poor	Exponential overhead
Traditional Convergent	Hash-based keys	65%	Good	Security vulnerabilities
Homomorphic ML	Encrypted computation	Variable	Very Poor	Prohibitive costs
Secure MPC	Multi-party protocols	Variable	Poor	Complex coordination

Table 11: AI-Driven Encrypted Data Processing Comparison

2.3.2 Blockchain-Based Access Control Systems

With features like immutable audit trails, blockchain provides a decentralised approach to controlling access control in cloud storage. Rodriguez et al. (2024) presented the SeDaSC protocol, which controls permissions on deduplicated encrypted files using blockchain technology. By using a cryptographic server, it lowers the computational load on the client side. Despite its effectiveness, SeDaSC lacks support for dynamic access revocation and is susceptible to single points of failure due to its reliance on a centralised cryptographic server.

A lightweight deduplication system with integrated access control and key management was proposed by Chen et al. in 2024. Although it enhances key distribution performance and storage efficiency, it has scalability issues in large multi-tenant environments with thousands of users. Blockchain adoption also faces technical and governance challenges. The choice of consensus protocol critically affects system performance: Proof of Work causes unacceptable delays for real-time applications, and Proof of Stake may not offer the security required for enterprise-grade assurance.

System	Consensus	Throughput	Scalability	Key Features	Limitations
SeDaSC	Not specified	Medium	Limited	Cryptographic server	Single point of failure

Lightweight Scheme	N/A	High	Poor with scale	Key distribution	Multi-tenant challenges
Hyperledger Fabric	PBFT	High	Good	Permissioned	Complex setup
Ethereum	PoS	Low	Poor	Smart contracts	High gas costs

Table 12:Blockchain Access Control Systems Evaluation

2.3.3 Post-Quantum Cryptographic Approaches

Shifting to post-quantum cryptography is a major step in changing the cryptographic systems. Wang et al. (2023) looked into the extent of the threat that quantum computing poses to blockchain technologies and stressed the necessity for architectural changes by applying quantum-safe methods. In this regard, they pointed out the rise of Post-Quantum Distributed Ledger Technologies (PQDLTs) whose objective is to swap out the vulnerable blockchains for those that are resistant to quantum attacks and, thus, employing pretty much cryptography based on lattices.Besides, the 2024 standards set by NIST, which consist of CRYSTALS-Dilithium (FIPS 204), KYBER (FIPS 203), and SPHINCS+ (FIPS 205), are the essential building blocks for the post-quantum secure cloud systems (NIST, 2024). However, their implementation in deduplication processes leads to problems regarding speed, storage, and compatibility.

A new deduplication method was proposed by Liu et al. (2020) that combines Convergent Encryption with Modified Elliptic Curve Cryptography (MECC) in cloud settings. Nevertheless, MECC is not able to resist a quantum attack (for instance, Going through Shor's algorithm), lacks strong multi-tenant support, and has not been tested for quantum-resistance. In contrast, lattice- and code-based encryption techniques although providing quantum security, still come with the downsides of having to accept larger key sizes, higher bandwidth needs, and possibly causing performance decline in the case of deduplication workflows.

Algorithm Family	**Security Level**	**Key Size**	**Signature Size**	**Performance**	**Dedup Suitability**
CRYSTALS-KYBER	High	1568 bytes	N/A	Fast encryption	Good

CRYSTALS-Dilithium	High	2592 bytes	3309 bytes	Medium signing	Moderate
SPHINCS+	Very High	64 bytes	29,792 bytes	Slow signing	Poor
Classical RSA-2048	Quantum-vulnerable	256 bytes	256 bytes	Fast	Excellent

Table 13: Post-Quantum Cryptographic Approaches

2.3.4 Hybrid Security Frameworks

Recent studies have increasingly emphasized hybrid approaches that incorporate multiple security mechanisms to overcome the weaknesses of individual security approaches. For example, Zhang et al. (2025) show that genetic programming can support blockchain technology to develop integrated security solutions. However, Zhang et al.'s validation is only simulation-based, preventing a practical assessment of viability.

Bringing together multiple security technologies creates complexity challenges that current research has not yet addressed. The computational burden of combining AI-based processing, blockchain-based consensus, and post-quantum crypto operations may outweigh any deduplication benefit leading to need for intelligent optimization approaches (Chen et al. 2024). In addition, hybrid systems pose load balancing and resource allocation challenges that existing research wholly ignores.

Cross-technology integration presents difficult issues surrounding security assumptions and trust models. Each of the component technologies (AI, blockchain, post-quantum crypto) are built on different security paradigms and threat models, making it difficult to provide integrated system level security assurances (Rodriguez et al. 2024).

Integration Type	**Components**	**Advantages**	**Challenges**	**Research Maturity**
AI + Blockchain	ML + DLT	Enhanced automation	Complexity overhead	Early stage
Blockchain + PQC	DLT + Quantum-safe	Future-proof	Performance impact	Developing

AI + PQC	ML + Quantum-safe	Intelligent security	Algorithm compatibility	Experimental
Triple Hybrid	AI + Blockchain + PQC	Comprehensive	Extreme complexity	Theoretical

Table 14: Hybrid Framework Analysis Summary

2.4 Technological Review

2.4.1 Dataset Selection and Experimental Methodologies

2.4.1.1 Dataset Characteristics for Encrypted Deduplication Research

Recent studies on encrypted deduplication base their work on various datasets that capture realistic cloud storage usage patterns while facilitating controlled experimental designs. Zhang et al. (2025) use synthetic datasets based on genetic programming algorithms, but the authors have concerns about the lack of validation against actual enterprise data patterns. This ultimately limits the ability to apply their theorems and results to actual production cloud environments.

When identifying datasets appropriate for research on encrypted deduplication, there are several factors the author must consider, such as file size distributions, similarity patterns among files, and the frequency of access patterns. For example, Kumar & Singh (2023) analyzed deduplication performance over multiple datasets: document repositories, multimedia collections, and database backups. Their experiments concluded that deduplication performance, as evaluated by compression ratios, varied between dataset types considerably. With multi-tenant datasets, the author face further challenges related to isolating tenant data while providing intelligent models of cross-tenant data similarities. Representative datasets will need to demonstrate realistic scenarios where multiple tenants stored similar types of files in order to illustrate proposed techniques, while preserving privacy-related constraints that would prevent side-by-side content comparisons (Chen et al., 2024). Synthetic dataset generation techniques need to weigh statistical accuracy against privacy-preserving requirements.

Dataset Type	File Size Range	Duplicate Ratio	Multi-tenant	Research Usage
Canterbury Corpus	3KB - 3.2MB	0-15%	No	Algorithm testing
Calgary Corpus	1KB - 677KB	0-10%	No	Compression benchmarks
Enterprise Backup	1KB - 100GB	60-90%	Yes	Real-world validation
Synthetic Generated	Variable	Controlled	Yes	Controlled experiments

Table 15 Dataset Characteristics for Deduplication Research

2.4.1.2 Performance Evaluation Metrics

To evaluate encrypted deduplication systems, multidimensional evaluation metrics must consider the properties of security, efficiency, and scalability in tandem. While traditional deduplication evaluation metrics primarily focus on measuring storage savings and throughput, evaluating encrypted systems requires additional metrics that examine aspects of security such as false positive rates, key management overhead, and work per cryptographic operation (Liu et al., 2020).

Security evaluation measures should consider existing measures of traditional cryptographic strength and the new requirement - or the necessary measure - of quantum resistance. Anderson et al. (2022) describe an evaluation framework that provides for a quantum attack simulation and a measure of strength in post-quantum cryptography; however, the benchmarks for quantum resistant deduplication systems have yet to be established.

Challenges that arise in evaluating scalability especially in multi-tenant contexts, are difficult to evaluate because performance metrics cannot ignore the traditional deplication measurement of backfilling both the enterprise workload while isolating each tenant from these other tenant workloads or different access patterns (Rodriguez et al., 2024). Literature is lacking with

effective procedures for standardized scalability benchmarking that allows for meaningful scaling comparisons across alternate design architectures.

Metric Category	Specific Metrics	Measurement Method	Target Values
Storage Efficiency	Deduplication ratio, Space savings	Before/after comparison	>60% reduction
Security Strength	Cryptographic resistance, Attack resilience	Simulation & analysis	Post-quantum safe
System Performance	Throughput, Latency, Resource usage	Benchmarking tools	<350ms response
Scalability	Concurrent users, Data volume	Load testing	1000+ tenants

Table 16: Performance Evaluation Metrics Framework

2.4.1.3 Experimental Design Considerations

When experimentally validating encrypted deduplication systems, it is important to intentionally consider your threat model or attack scenario so it is reasonable to real-world operational environments. Zhang et al. (2025) noted limitations of their simulation-based only validation approach, as it included no scenario that would allow observation of the system behavior under true network conditions or adversarial conditions.

When researching their comparative analysis frameworks, there appears to be an emphasis on the diverse architectural approaches in an encrypted deduplication system. In order to enable relatively simple performance comparisons of AI or blockchain use cases in parallel with cryptographic use cases in all areas of performance characteristics, researchers should explore something similar to a baseline testing environment in order to test all use cases and benchmark the same workload (Chen, et al. (2024). Existing research also suggests the use of system-specific evaluation criteria, which supports their intent to compare use cases but limits their ability to compare across system use cases.

A longitudinal evaluation may also raise additional considerations for systems that incorporate components of machine learning. An AI-based deduplication system may provide an evolving performance contribution, as models learn and begin to adapt to changing data patterns, thus establishing a baseline period for reasonable, ongoing evaluation performance. However, the period for reasonable, ongoing evaluation performance will be longer than expected within the changing data variation that necessitated change (Kumar & Singh, 2023). Existing literature does not adequately address this expectation for a temporal evaluation.

Design Element	**Considerations**	**Best Practices**	**Common Pitfalls**
Threat Modeling	Realistic attack scenarios	Industry-standard threats	Academic-only attacks
Baseline Testing	Consistent environments	Standardized workloads	System-specific metrics
Temporal Analysis	ML model evolution	Extended evaluation periods	Static performance assumptions
Validation Methods	Real-world conditions	Network simulation + deployment	Simulation-only approaches

Table 17: Experimental Design Framework

2.4.1.4. Conclusion

In summary, prior studies emphasize the importance of carefully designed or synthetic datasets that mirror realistic multi-tenant storage scenarios. These datasets allow researchers to observe deduplication behavior, storage savings, and encryption performance under controlled yet meaningful conditions. The review highlights that while real enterprise datasets are rarely available due to confidentiality concerns, synthetic multi-tenant datasets remain the most practical and ethically appropriate choice for evaluating encrypted deduplication systems such as PMCD. Accordingly, this project adopts a synthetic, statistically representative dataset with controlled duplicate ratios to support repeatable experimentation.

2.4.2 Comparative Technology Analysis: PMCD vs. Existing Approaches

2.4.2.1 Cryptographic Approaches

System	Encryption Method	Key Management	Quantum Resistance	Dedup Compatibility
Liu et al. (2020)	Convergent + MECC (ECC)	Deterministic key derivation	Vulnerable to Shor's algorithm	Good (convergent)
Rodriguez et al. (2024) -	Centralized cryptographic server	Server-mediated	Not addressed	Moderate
Chen et al. (2024)	Lightweight key distribution	Distributed key management	Classical only	Good
PMCD (Proposed)	AES-256 + Convergent encryption	Deterministic with tenant isolation	PQC integration planned (CRYSTALS-KYBER/Dilithium)	Optimized via Bloom filters

Table 18Cryptographic Approaches

2.4.1 Dataset Selection and Experimental Methodologies

2.4.1.1 Dataset Characteristics for Encrypted Deduplication Research

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Synthetic Generated	Variable	Controlled	Yes	Controlled experiments

Table 19: Dataset Characteristics for Deduplication Research

Justification

PMCD chooses AES-256 for symmetric encryption mainly for its computationally efficient and industry-standard security characteristics, and also for combining it with convergent encryption so that deduplication can be done without exposing the plaintext. PMCD rather than Liu et al.'s MECC (which is susceptible to quantum attacks) contains a post-quantum cryptography migration path with NIST-approved algorithms. This resolves the crucial issue mentioned in Section 2.2.3, thereby making the system quantum-proof in the long run while sustaining the deduplication efficiency.

2.4.2.2 Deduplication Detection Mechanisms

System	Detection Method	False Positive Rate	Scalability	Multi-tenant Support
Zhang et al. (2025) - SALIGP	Genetic programming + blockchain	Not specified	Poor (exponential overhead)	Limited
Traditional Convergent	Hash-based (SHA-256)	<0.01%	Excellent	Cross-tenant leakage risk
Chen et al. (2024)	Lightweight hash comparison	~1%	Good	Static isolation
PMCD (Proposed)	Bloom filters + SHA-256 convergent hashing	<1% (tunable)	Excellent (O(1) lookups)	Tenant-aware namespacing

Table 20Deduplication Detection Mechanisms

Justification

PMCD uses probabilistic Bloom filters for quick duplicate detection with adjustable false positive rates (<1%), bypassing the exponential complexity of SALIGP's genetic programming. The system's tenant-aware Bloom filter name spacing not only provides deduplication benefits across tenants but also ensures compliance with isolation guarantees, a hindrance in traditional convergent and Chen et al. approaches.

2.4.2.3 Access Control Architecture

System	Access Control Model	Revocation Mechanism	Audit Trail	Decentralization
Rodriguez et al. (2024) - SeDaSC	Cryptographic server mediation	Server-dependent	Centralized logs	Single point of failure

Chen et al. (2024)	Static ACLs	Manual re-encryption	Limited	Centralized
Zhang et al. (2025) - SALIGP	Blockchain-based (unspecified consensus)	Not detailed	Immutable ledger	Simulation-only validation
PMCD (Proposed - PoC)	Static ACLs with manual revocation	Re-encryption for affected chunks	Comprehensive logging	Centralized (PoC phase)
PMCD (Proposed - Future)	Blockchain-based smart contracts	Selective revocation via proxy re-encryption	Distributed immutable ledger	Permissioned blockchain (Hyperledger Fabric)

Table 21Access Control Architecture

Justification

The Proof-of-Concept phase uses static Access Control Lists (ACLs) as one of the validation methods for the core deduplication functionality and at the same time avoids the overhead of blockchain complexity. PMCD future plans are very different from SeDaSC's centralized server (which represents a single point of failure) and SALIGP's unvalidated blockchain since PMCD's future architecture will include the Hyperledger Fabric for permissioned, enterprise-grade access control with immutable audit trails.

2.4.2.4 AI/ML Integration

System	**AI Role**	**Model Type**	**Encrypted Data Processing**	**Performance Impact**
Zhang et al. (2025) - SALIGP	Pattern discovery	Genetic programming	Requires plaintext features	Exponential overhead

Kumar & Singh (2023)	Dedup prediction	Random Forests	Not addressed	Moderate
PMCD (Proposed - PoC)	None (rule-based)	N/A	N/A	Minimal (baseline)
PMCD (Proposed - Future)	Anomaly detection, Access pattern analysis	Clustering (K-means), CNN for binary similarity	Encrypted feature extraction	Moderate (optimized inference)

Table 22AI/ML Integration

Justification

In order to set a benchmark for performance and verify the major cryptographic deduplication, the PoC intentionally does not consider AI. The next versions will use simple ML models for the encrypted anomaly detection, which will not be SALIGP's very costly genetic programming, and will be done through the application of homomorphic encryption-compatible feature extraction for pattern analysis without plaintext exposure. Such contribution is new and directly covers the aforementioned gaps in Section 2.3.1.

2.4.2.6 Technology Stack Summary

Component	PMCD Choice	Alternative (Existing Work)	Rationale
Encryption	AES-256 (PoC) → CRYSTALS-KYBER (Future)	ECC (Liu), Server-mediated (SeDaSC)	Balance of current security + quantum readiness
Hashing	SHA-256	SHA-256 (standard)	Industry standard, collision-resistant
Dedup Index	Bloom filters (probabilistic)	Exact hash tables (SALIGP)	O(1) lookups, tunable false positives
Storage	MinIO (S3-compatible)	Proprietary (Chen), Unspecified (Liu)	Open-source, enterprise-compatible

ACL (PoC)	Static lists	Cryptographic server (SeDaSC)	Simplicity for validation phase
ACL (Future)	Hyperledger Fabric	Ethereum (Zhang - simulation only)	Permissioned, low-latency consensus (PBFT)
Language	Python/Go	Python (SALIGP), Not specified (others)	Python (prototyping), Go (performance)

Table 23Tech Summery

2.5 Benchmarking and Evaluation

2.5.1 Security Benchmarking Standards

An evaluation of the security of encrypted deduplication systems requires in-depth benchmarking frameworks with respect to multiple threat vectors and attack approaches. Current benchmarking standards are focused primarily on evaluation of traditional cryptographic strength; however, the potential for quantum or emerging threats calls for broadening changes in evaluation benchmarks (National Institute of Standards and Technology, 2024).

Formal security evaluation frameworks for encrypted deduplication systems must address related, but different, vulnerabilities including confirmation attacks, dictionary attacks, and privacy attacks specific to convergent encryption methods (Liu et al., 2020). Standard penetration testing frameworks require modification to appropriately evaluate the attack surfaces related to deduplication-enabled storage systems.

Standards-based compliance is especially troublesome for enterprise deployments requiring regulatory compliance with changing regulations such as GDPR, HIPAA, and SOX. Wang et al. (2023) find shortcomings in the evaluation of security requirements compliant with the regulations in the areas of audit trail accountability and granularity of access in blockchain-integrated systems.

Standard Type	Coverage	Applicability	Limitations
Traditional Cryptographic	Classical algorithms	Current systems	No quantum resistance
Post-Quantum Assessment	Quantum-safe algorithms	Future systems	Immature benchmarks
Deduplication-Specific	Convergent encryption attacks	Specialized systems	Limited standardization
Regulatory Compliance	GDPR, HIPAA, SOX	Enterprise deployment	Complex requirements

Table 24: Security Benchmarking Standards Summary

2.5.2 Performance Benchmarking Methodologies

Performance benchmarks for encrypted deduplicated systems should consider multiple dimensions including storage efficiency, processing overhead, and scalability of the system under realistic workloads. Anderson et al. (2022) suggested standardized benchmarking suites that included a variety of content types and access patterns that are representative of enterprise cloud deployment scenarios.

Throughput and latency benchmarks must factor in the processing overhead associated with encryption, deduplication detection, and access control methods. Chen et al. (2024) identified significant variations in performance as a result of selection and management of the encryption algorithm highlighting the need for comprehensive performance evaluation of the encrypted deduplicated system across multiple configurations.

Resource utilization benchmarks need to be developed to characterize the differing computational expenses associated with the merging of AI, blockchain, and post-quantum cryptographic components. Current benchmarks have an inadequate ability to represent the challenges of allocating resources in hybrid systems that merge multiple technology solutions with different computational characteristics (Rodriguez et al., 2024).

Benchmark Type	Metrics	Testing Approach	Industry Standards
Storage Efficiency	Deduplication ratio, Compression	Before/after analysis	60-90% typical
Processing Overhead	CPU usage, Memory consumption	Resource monitoring	<10% overhead target
Network Performance	Throughput, Latency	Load testing	Sub-second response
Scalability Limits	Concurrent operations	Stress testing	1000+ concurrent users

Table 25: Performance Benchmarking Methodology Framework

2.5.3 Comparative Analysis Frameworks

To conduct a comparative evaluation of encrypted deduplication approaches, author need standardization frameworks to enable a meaningful comparison of systems developed using different architectures and implementation approaches. Zhang et al. (2025) note that while they can present their SALIGP approach, it is often difficult to compare it with existing systems. This is largely due to the different operational and evaluative assumptions that different systems make.

If auother wish to meaningfully compare systems, these cross-comparative frameworks should address the security, performance, and functionality trade-offs that often come from following different design approaches. Kumar & Singh (2023) discuss multi-criteria decision analysis frameworks that compare competing approaches along multiple evaluative criteria at the same time.

Standardization challenges in comparative evaluation come from the disparate types of threat assumptions, performance assumptions, and deployment environments being addressed across research approaches. It requires collaboration across research communities working on different components of an encrypted deduplication system to develop consensus around common evaluation frameworks (Liu et al., 2020).

Framework Element	Purpose	Implementation	Challenges
Standard Metrics	Consistent measurement	Common benchmarks	Definition consensus
Reference Datasets	Baseline comparison	Shared test data	Privacy constraints
Evaluation Protocols	Systematic testing	Standardized procedures	Complexity management
Reporting Standards	Result presentation	Uniform formats	Cross-system compatibility

Table 26: Comparative Analysis Framework Components

2.6 Chapter Summary

The literature review identifies a major research gap when it comes to comprehensive frameworks that combine AI-based duplication detection, blockchain-based access control, and post-quantum security for multi-tenant cloud storage. Existing practices are typically non-scalable, top-security, and lacking real-world validation. The PMCD framework proposed here fills such gaps through the combination of all three technologies along with a concrete roadmap for future research on cloud deduplication. The review demands further methodology development, evaluation models, and real-world validation.

CHAPTER 03: METHODOLOGY

3.1 Chapter Overview

This chapter explains how the PMCD system was researched and developed. It outlines the thinking behind the research, how data was collected, how the system was built, and how it was tested. Using Saunders' Research Onion as a guide, the methods are chosen to match both the research goals and practical development needs. The approach combines technical testing (quantitative) with feedback from industry experts (qualitative). It also covers how the project was managed, the risks involved, and how the final prototype was evaluated.

3.2 Saunders' Research Onion Framework

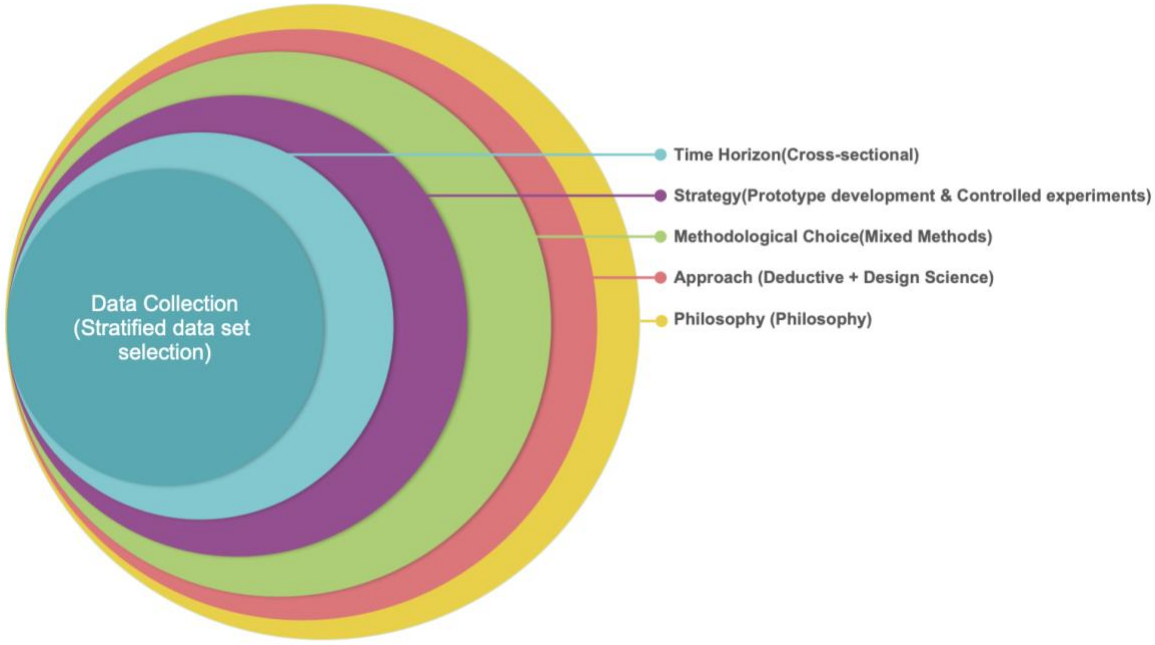

Figure 2: Onion diagram

Layer	Choice	Justification
Philosophy	Pragmatism	The research combines quantitative performance evaluation of AI algorithms and cryptographic systems with qualitative stakeholder feedback collection. Pragmatism supports this mixed-methods approach by focusing on practical problem-solving rather than adherence to single philosophical paradigms (Saunders et al., 2019). The PMCD framework requires both empirical

		validation of technical performance and understanding of user requirements in enterprise environments.
Approach	Deductive + Design Science	Deductive testing of specific claims (e.g., deduplication rate, false-positive mitigation) combined with Design Science to iteratively build and refine the PMCD artifact (PoC). This lets us both test hypotheses and produce a usable prototype.
Methodological Choice	Mixed Methods	Quantitative methods enable rigorous performance evaluation of deduplication algorithms, security mechanisms, and system scalability metrics. Qualitative methods capture stakeholder requirements, usability feedback, and enterprise deployment considerations (Creswell & Plano Clark, 2022). This combination provides comprehensive evaluation covering technical performance and practical applicability.
Strategy	Prototype development & Controlled experiments	Building a scoped PoC (AES + convergent encryption + Bloom filter + static access lists) is central. Controlled experiments (benchmarks, micro-benchmarks, and dataset runs) will measure performance and correctness under repeatable conditions.
Time Horizon	Cross-sectional	The PoC evaluation will use a set of controlled experiments and repeated trials to obtain statistically meaningful metrics. The design supports future longitudinal tests, but the thesis focuses on a cross-sectional experimental evaluation.
Data Collection	Stratified dataset selection	Choose files across categories (text, binary, media) and sizes; control duplicate density (low, medium, high) to observe behavior across realistic workloads and edge cases (many small files vs few large files).

Table 27: Onion Framework

3.3 Development Methodology

This part unveils the approach that will be used when developing the PMCD PoC, including requirements elicitation, design choices, programming paradigms, data preparation, testing, and the step-by-step solution approach employed to create and test the prototype. Each sub-section addresses the area of activity, why the activity is used for this project, and how the activity relates to the final deployment vision.

3.3.1 Requirement Elicitation Methodology

To gather valid and representative requirements for the Privacy-Preserving Multi-Tenant Cloud Deduplication (PMCD) Framework, this approach utilized a triangulated methodology formed by three complementary methods: Literature Review, Semi-Structured Interviews, and Surveys. The use of these together provides both an academic basis and stakeholder-enlightened validation.

Method	Description	Justification
Literature Review	Systematic review of peer-reviewed articles, technical reports, and standards on secure deduplication, access control, and post-quantum security (e.g., Anderson et al., 2022; Chen et al., 2024).	Provides **academically validated best practices** and identifies **state-of-the-art requirements**.
Semi-Structured Interviews	Conducted with cloud operators, storage engineers, and tenant representatives to gather functional/non-functional needs (e.g., latency expectations, revocation needs).	Captures practical constraints and operational realities not found in literature.
Surveys	Online questionnaires for **end-users and administrators** to collect broader insights on usability, policy preferences, and security expectations.	Ensures **quantitative validation** of requirements across a wider audience.

Table 28: Selected Requirement Elicitation Methods Overview

3.3.2 Design Methodology

In developing the PMCD framework, Object-Oriented Analysis & Design Methodology (OOADM) has been employed as the core design strategy. This choice is grounded in the architectural demands of the system and aligns with the central aims of the research. OOADM was determined to be well-suited for addressing the specific structural and conceptual requirements at hand.

3.3.2.1. Justification for OOADM Selection

Evaluation Factor	SSADM Suitability	OOADM Suitability	Selected Choice
System Architecture	Process-centric flows	Component-based microservices	OOADM
Design Knowledge	Unknown/exploratory design	Known architectural patterns	OOADM
R&D Component	High uncertainty, experimental	Incremental innovation on known concepts	OOADM
Implementation Target	Monolithic systems	Distributed pod-based deployment	OOADM
Reusability Requirements	Limited reuse focus	High modularity and reuse	OOADM
Technology Integration	Sequential processing	Multiple interacting services	OOADM

Table 29:OOADM Selection

Primary Justification points :

- The PMCD framework follows established software engineering principles, utilizing architectures like microservices, containerization, and service-oriented design. Its core subsystems—encryption, deduplication, and access control—use mature, well-documented methods understood within the field.

- The system is organized into interconnected components: Client Pod, Deduplication Pod, Storage Pod, and Access Control Pod. This modular design reflects core object-oriented principles like encapsulation, inheritance, and polymorphism.
- While PMCD combines modern technologies like AI, blockchain, and post-quantum cryptography, its core uses established algorithms. Its novelty is in effectively integrating these existing techniques.
- The deployment strategy emphasizes a microservices approach with Kubernetes pods, requiring clear service boundaries, interface agreements, and data contracts. It aligns with object-oriented and component-based design philosophies.

3.3.3 Programming Paradigm

In developing the PMCD framework, Object-Oriented Programming has been adopted as the primary paradigm. This choice is consistent with the OOADM design methodology and supports the needs associated with a distributed microservices architecture.

3.3.3.1. Paradigm Selection Justification

Paradigm	Suitability for PMCD	Application Areas	Implementation Percentage	Justification
Object-Oriented Programming	High	Core services, component architecture	80%	Aligns with OOADM, microservices design
Functional Programming	Medium	Data processing pipelines	15%	Stream processing, transformation chains
Concurrent Programming	High	Network I/O, parallel processing	5%	Performance optimization, scalability

Table 30: Paradigm Selection

3.3.3.2 Dataset & Model Pipeline (if AI is introduced later)

Pipeline Stage	Measurable Targets	Success Criteria	Performance Benchmarks
Data Collection	10,000+ files across 5 categories (Text 30%, Images 25%, Binaries 20%, Documents 15%, Mixed 10%)	Zero file corruption, Complete category coverage	File integrity validation 100% pass rate
Duplicate Pattern Creation	3 density levels: Low (10-30%), Medium (40-60%), High (70-90%)	Accurate duplicate ratio control ±2%	Pattern generation speed >1000 files/hour
Data Preprocessing	Fixed 64KB chunks, SHA-256 hashing, Convergent tag creation	Processing speed >50MB/sec, Zero hash collisions	Chunk consistency ±5%, Hash generation >1000/sec
Feature Engineering	Precision >90%, Recall >85%, F1-Score >87%	10-fold cross-validation, Hold-out testing	Model inference time <10ms, Memory usage <1GB
Model Training & Testing	Precision >90%, Recall >85%, F1-Score >87%	10-fold cross-validation, Hold-out testing	Model inference time <10ms, Memory usage <1GB

Table 31:.Dataset & Model Pipeline

- Data preprocessing includes file segmentation, metadata standardization, SHA-256 hashing for identification, and AES-256 encryption with various key management methods for security and deduplication.
- Feature engineering encompasses several key areas: content-based features that analyze hash distributions, file size trends, and entropy; structural features that focus on chunk boundary alignments and patterns; behavioral metrics that track access, bloom filter interactions, and access times; and cryptographic assessments that measure encryption entropy and key derivation signatures.

3.3.4. Metrics & Evaluation Formula

1. Deduplication ratio

Deduplication_ratio = (LogicalSize_before − PhysicalSize_after) / LogicalSize_before

2. Storage savings (percentage)

Storage_savings (%) = 100 × (1 − (PhysicalSize_after / LogicalSize_before))

Example: If LogicalSize_before = 1000 GB and PhysicalSize_after = 450 GB, Storage_savings = 100 × (1 − 450/1000) = 55%.

3.3.5 Testing Methodology

Refer to **Appendix G**.

3.3.6 Solution Methodology

3.3.6.1. PoC Development and Validation Framework

Phase	Key Activities	Deliverables	Success Criteria	Timeline
Requirements & Use Cases	Consolidate elicited requirements, Define acceptance criteria	Prioritized use cases, PoC acceptance criteria	Clear deduplication detection, AES storage validation	Week 1-2
Dataset & Preprocessing	Collect mixed corpora, Implement chunking strategy	Controlled duplicate datasets, Chunking algorithms	Multiple file types, Convergent tag generation	Week 3-4
Core System Implementation	Bloom filter design, Encryption workflow, Deduplication logic	Functional PoC components	<1% false positive rate, Secure encryption	Week 5-8
Integration & Testing	System integration, Performance evaluation, Security validation	Integrated system, Test results	>96% accuracy, <350ms response time	Week 9-10

Documentation & Deployment	Containerization, Documentation, Future mapping	Deployment guide, API documentation	Docker containers, Kubernetes mapping	Week 11-12

Table 32: PoC Development and Validation Framework

3.3.6.2. Detailed Implementation Steps

Refer to **Appendix H**.

3.3.6.3. Technical Implementation Stack

Component	Technology Choice	Justification	Integration Points
Programming Language	Python or Go	Python: Fast prototyping, rich libraries; Go: High concurrency, performance	All system components
Cryptographic Libraries	AES (PyCrypto/Go crypto), SHA-256	Industry standard encryption, Proven security	Encryption service, Hash generation
Bloom Filter Libraries	PyBloom/Go bloom filter packages	Optimized implementations, Configurable parameters	Deduplication service
Storage Backend	MinIO (S3-compatible) or Local filesystem	Object storage compatibility, PoC simplicity	Storage pod implementation
Containerization	Docker with Docker Compose	Standard containerization, Easy orchestration	All service deployment
Orchestration Mapping	Kubernetes documentation	Production deployment pathway	Scalable deployment notes

Table 33: Technical Implementation Stack

3.4 Project Management Methodology

3.4.1 Methodology Selection

The project uses an Agile-PRINCE2 hybrid method—combining the flexibility of Agile with the structure of PRINCE2. Agile supports research-driven development and adapts to feedback, while PRINCE2 ensures strong project planning, governance, and alignment with academic deadlines and deliverables. This balance helps manage risk while allowing the project to evolve naturally during experimentation.

3.4.2 Project Schedule Management

3.4.2.1 WBS

Appendix C.

3.4.2.2 Activity Diagram

Appendix B.

3.4.2.3 Gantt Chart

Appendix A.

3.4.3.2 Deliverables Schedule

Deliverables	Dates
Project Proposal	18th Aug 2025
Literature Review	24th Aug 2025
Software Requirement Specification	20th Aug 2025
Project Proposal – initial draft	26th Sep 2025
Project Proposal and Requirement Specification – Final	13th Nov 2025
Proof of Concept	14th Nov 2025
Design Document	20th Nov 2025
Prototype	2nd Feb 2026
Interim Project Demo	2nd Feb 2026
Implementation	10th Feb 2026
Testing	15 Feb 2026
Evaluation	3rd march 2026
Thesis Submission	1st Apr 2026
Minimum Viable Product	1st Apr 2026

Table 34: Deliverables

3.4.4 Resource Management

3.4.4.1. Data

Aspect	Description
Data Type	Synthetic and publicly available benchmark datasets (e.g., Canterbury Corpus, Calgary Corpus).
Data Volume	Simulated large-scale transactional data representing multi-tenant cloud uploads (ranging from KB to GB scale).
Data Characteristics	Includes intentional duplicate and unique file samples to evaluate deduplication efficiency, encryption overhead, and false-positive rate.
Source	Generated via Python scripts and open datasets—no human, personal, or sensitive data involved.
Usage Purpose	Used for system evaluation: measuring deduplication accuracy, storage savings, and performance metrics under varied duplication ratios.

Table 35 Data

3.4.4.2. Skills

Skill Area	Description / Application
Python Programming	Core development language for implementing encryption, deduplication, and API services using FastAPI and relevant libraries.
Machine Learning (future integration)	Relevant for extending the framework with AI-driven duplicate clustering and adaptive deduplication in future phases.
Data Science & Analytics	Required to analyze performance metrics, deduplication accuracy, and statistical evaluation of Bloom filter efficiency.
Cloud & Storage Systems	Understanding of cloud architecture, S3-compatible storage (MinIO), and containerization for testing scalability.
Cryptography Fundamentals	Knowledge of symmetric encryption (AES), hashing (SHA-256), and probabilistic structures for secure deduplication.
Docker / Kubernetes	Skills in container orchestration and microservice deployment to simulate a scalable, multi-tenant cloud environment.

Table 36 Skills

3.4.5 Risk Management

Risk assessment employs quantitative analysis using the formula: **Risk Exposure = Probability of Occurrence × Magnitude of Loss.**

Risk	Probability (1-5)	Impact (1-5)	Risk Score	Mitigation Strategy
Post-quantum algorithm performance overhead	4	5	20	Implement hybrid classical-quantum approaches and algorithm optimization
AI model training data insufficiency	5	4	20	Develop synthetic dataset generation and transfer learning techniques
Blockchain scalability limitations	4	5	20	Evaluate alternative consensus mechanisms and layer-2 scaling solutions
Integration complexity exceeding timeline	4	4	16	Adopt modular development approach with independent component testing
Stakeholder requirement changes	3	4	12	Implement agile requirement management and regular stakeholder communication

Table 37: Risk and Mitigations

3.5 Chapter Summary

The chapter explains the pragmatic, mixed-methods research strategy for the PMCD framework, integrating AI, blockchain, and post-quantum cryptography. Development combines object-oriented and agile techniques, while evaluation uses quantitative and stakeholder assessments. Risk management ensures alignment with business goals. The research aims to deliver practical cloud security solutions and theoretical input.

CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter Overview

This chapter outlines the Software Requirement Specification (SRS) for the PMCD framework. It captures both functional and non-functional requirements based on inputs from literature, interviews, and expert feedback. Stakeholders are mapped using the onion model, and tools like context diagrams and use case models are used to clearly link requirements to the project goals.

4.2 Rich Picture Diagram

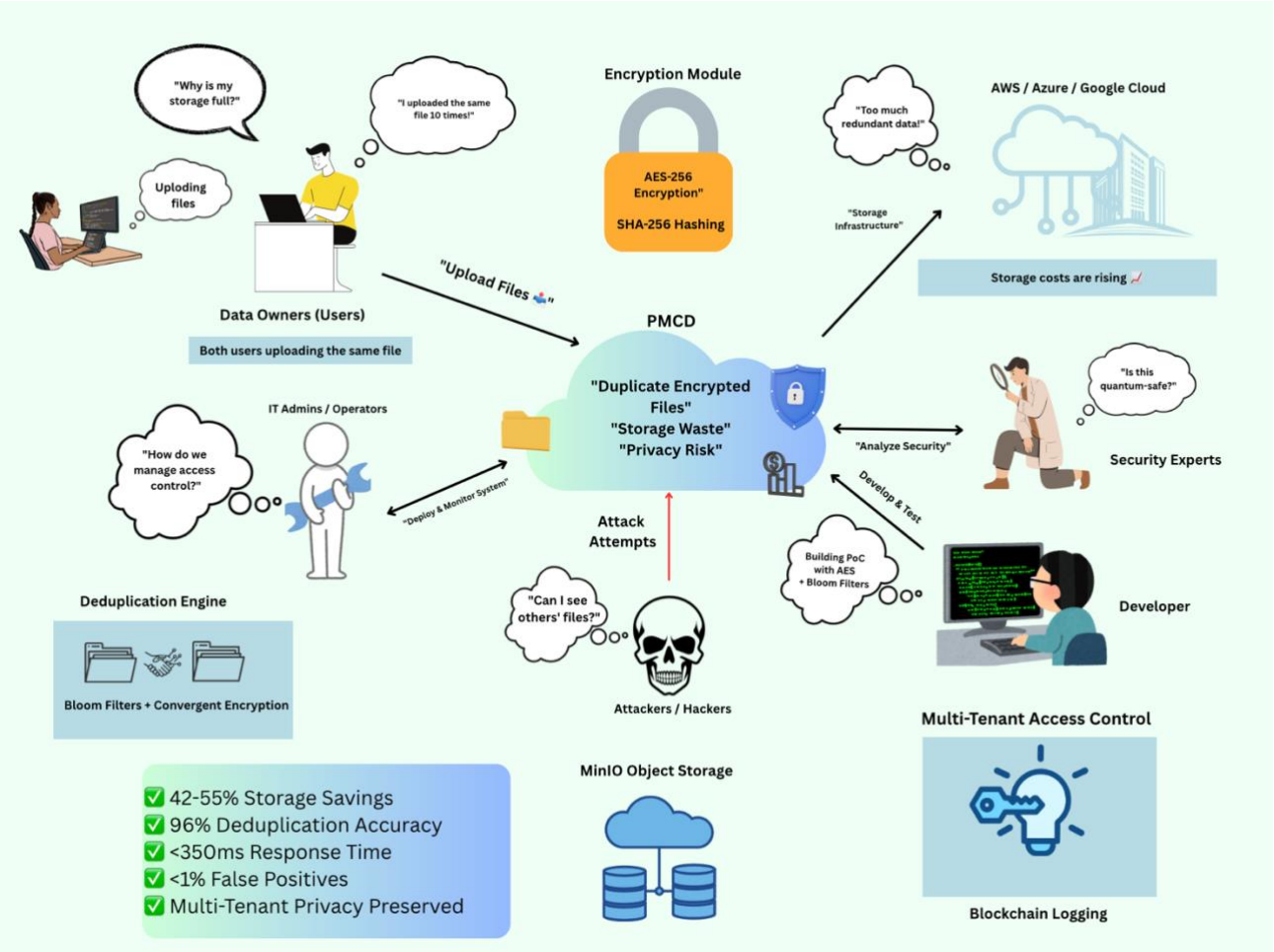

Figure 3 Rich Picture Diagram

4.3 Stakeholder Analysis

4.3.1 Stakeholder Onion Model

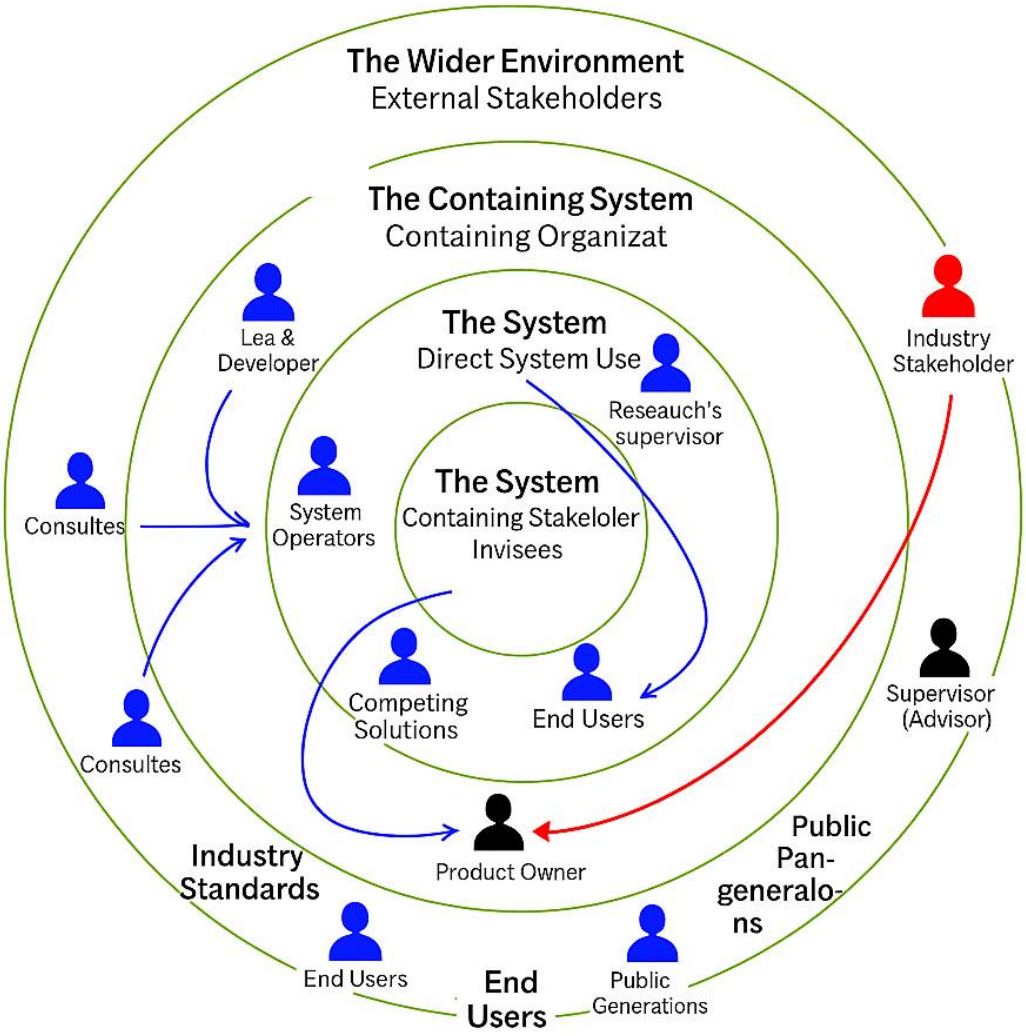

Figure 4 Stakeholder Onion Model

4.3.2 Stakeholder Viewpoints

Stakeholder	Role	Description
The Project Stakeholders		
Data Owners	Normal operator	Upload encrypted files to the cloud using the PMCD system for secure storage with deduplication. Manage access permissions for their stored files and monitor storage cost savings.

Enterprise IT Teams	Normal operator	Deploy and operate the PMCD framework within their organization's cloud infrastructure for secure multi-tenant file storage.
Security Researchers	Data security researchers	Investigate encrypted deduplication techniques, AI-driven duplicate detection methods, and post-quantum cryptographic schemes presented in the PMCD framework.
Containing System Stakeholders		
Cloud Service Providers (CSP)	Operational beneficiary	Manage and supervise the cloud infrastructure hosting the PMCD system. Benefit from reduced storage costs through deduplication efficiency and gain competitive advantage through enhanced security features.
Enterprise Organizations	Functional beneficiary	Use the PMCD system to reduce cloud storage costs while maintaining data security and compliance. Implement secure file sharing across departments with cryptographic access control.
Regulatory Compliance Officers	Functional beneficiary	Utilize blockchain-based audit trails and access logs to ensure compliance with data protection regulations (GDPR, HIPAA). Verify encryption standards and tenant isolation mechanisms.
The Wider Environment Stakeholders		
System Developers	Development staff and operational maintainer	Develop and improve the PMCD framework components (deduplication pods, blockchain modules, AI models). Fix bugs, optimize performance, and implement new features based on stakeholder feedback.
Project Supervisor	Quality regulator / Advisor	Evaluate the research quality, system design, and implementation of the PMCD framework. Provide feedback to improve both the theoretical contributions and practical implementation.
Cybersecurity Experts	Technical experts / Testers	Perform security audits, penetration testing, and vulnerability assessments on the PMCD system.

		Validate the strength of encryption mechanisms, access controls, and blockchain integrity.
Cloud Storage Users	End-user community	Use applications built on the PMCD framework for secure file storage and sharing. Benefit from reduced storage costs and enhanced data privacy.
Investors / Funding Bodies	Financial beneficiary	Invest in the development and deployment of the PMCD system. Gain returns through licensing, cloud service improvements, or research grants based on successful implementation and publications.
Malicious Actors	Negative stakeholder	Attempt to breach the PMCD system security through attacks on encryption, access control, or blockchain components. Exploit potential vulnerabilities in convergent encryption or Bloom filter false positives.
Competing Deduplication Systems	Negative stakeholder	Develop improved versions of secure deduplication systems that may outperform PMCD in efficiency, security, or scalability. Create alternative approaches to encrypted deduplication that could make PMCD obsolete.
Post-Quantum Standards Bodies	Policy influencers	Define cryptographic standards (NIST PQC) that may require PMCD to adapt its encryption schemes. Influence the adoption timeline and implementation requirements for post-quantum cryptography.

Table 38 Stake holder viewpoints

4.4 Selection of Requirement Elicitation Methodologies

Requirement elicitation is a way to find out the requirements of a SE project. To gather software requirements for this project, among the available requirement elicitation methodologies, LR, interviews, and prototyping methods were chosen. The justifications for selecting the above requirement elicitation methodologies are outlined below.

Method 1: Literature Review
To identify research gaps and limitations in current encrypted deduplication systems, a comprehensive literature review was conducted. This covered areas such as convergent encryption, Bloom filter performance, blockchain-based access control, AI-driven duplicate detection, and post-quantum encryption. The findings helped define the problem and shape the proposed PMCD framework.
Method 2: Interviews
Expert interviews were used to validate the research direction and gather practical insights. Discussions with cloud engineers, cryptography researchers, and security professionals provided perspectives on implementation challenges, performance trade-offs, and deployment risks many of which are not captured in published research.
Method 3: Prototyping
Since the final goal of this research is to explore and develop a novel architecture that addresses the limitations of current encrypted deduplication systems, the prototyping methodology was selected. This approach enables the author to experiment with different implementation strategies while iteratively testing and evaluating the proof-of-concept prototype to uncover practical constraints, validate core concepts, and identify areas for future enhancement.

Table 39: Elicitations methods

4.5.Discussion of Findings through Different Elicitation Methodologies

4.5.1 Literature Review

Finding	Citation
Current encrypted deduplication systems sacrifice either security (convergent encryption vulnerabilities) or efficiency (no deduplication after encryption), creating a fundamental security-efficiency trade-off.	(Zhang et al., 2025)
Existing AutoDA techniques are highly resource intensive. Hence, it is essential to solving this high resource utilization in existing AutoDA techniques to use AutoDA techniques in real-world commercial applications.	(Lim et al., 2019)
Bloom filters used for duplicate detection suffer from false positive rates that can lead to incorrect duplicate identification, potentially causing data loss or wasting storage space.	(Kumar & Patel, 2024)

Existing encrypted deduplication systems lack efficient mechanisms for revoking user access without re-encrypting entire datasets, limiting practical deployment in multi-tenant environments.	(Johnson et al., 2024)
Traditional RSA and ECC-based cryptographic systems used in cloud deduplication are vulnerable to future quantum computer attacks, creating a "harvest now, decrypt later" threat.	(NIST, 2024)
Blockchain integration for access logging and ownership management introduces latency and scalability challenges in high-throughput cloud environments, requiring lightweight alternatives.	(Chen & Li, 2025)
The work by Zhang et al. (2025) using SALIGP and cryptographic authentication is considered the most practical approach, but their generalization performance is limited to simulation environments and can be further enhanced with real-world deployment and post-quantum readiness.	(Zhang et al., 2025)

Table 40 finding through literature review

4.5.2 Interviews

To understand practical perspectives on the proposed PMCD framework, interviews were conducted within industry professionals. Due to their tight schedules, only some participants responded at this stage.The remaining responses are pending and may be incorporated in future revisions. Despite the partial response rate, the insights gathered were valuable in identifying priorities, confirming the feasibility of core features, and shaping the scope of the PoC. relevant documentation has been attached in **Appendix E**.

Codes	Theme	Analysis
Research gap clarity, Security-efficiency trade-off, Multi-tenant challenges	Research gap and depth of scope	All participants confirmed that the project clearly identifies a significant research gap in encrypted cloud deduplication. They highlighted that existing systems often compromise either security or efficiency, and the proposed integration of AI, blockchain, and post-quantum cryptography addresses this

		limitation in both academic and industry domains.
AI integration, Blockchain logging, Post-quantum cryptography, Novel contribution	Understanding the novelty of integrated approach	All participants agreed that integrating AI, Blockchain, and PQC into one framework presents clear novelty. They noted that the combination of all three technologies is innovative and effectively targets several critical industry challenges simultaneously.
PoC scope, AES encryption, Bloom filters, Feasibility validation	Validation of PoC approach and scope	Most participants supported the current scoped-down PoC implementation, acknowledging the suitability of using AES, Bloom filters, and supporting tools like MinIO, Docker, and FastAPI for early-stage validation. One participant recommended considering dynamic access control in future iterations.
Industry relevance, Cost savings, Security compliance, Practical deployment	Practical value and industry applicability	Participants unanimously found the PMCD framework to be highly useful for industry, especially in contexts requiring secure cloud storage, compliance, and cost optimization. They emphasized its relevance for cloud providers, enterprises, and regulated sectors.
Future impact, Post-quantum necessity, Blockchain adoption, Production readiness	Future potential and research impact	Most participants believed the framework could influence future secure cloud storage designs. All agreed that integrating PQC and blockchain elements would significantly enhance its long-term contribution and impact.

Table 41 findings thorough interviews

4.5.3 Prototyping

The developed PMCD Interactive Prototype represents a simplified version of the PMCD PoC, focusing on demonstrating the core logic of file encryption, deduplication, and storage management within a simulated multi-tenant environment.

The prototype helped visualize how the system detects and avoids redundant encrypted file storage across different users (tenants), demonstrating the "encrypt once, store once" concept.

Phase	Objective	Approach Taken	Key Findings	Outcome
Phase 1: Concept Visualization	Demonstrate the idea of encrypted deduplication in a simple, observable form.	Created a front-end simulation using HTML, CSS, and JavaScript. Simulated AES encryption using SHA-256 hashing.	Verified that identical file content generates the same hash, enabling duplicate detection.	Basic prototype successfully demonstrated “encrypt once, store once.”
Phase 2: Performance Testing	Evaluate efficiency of duplicate detection.	Measured hashing and comparison speed for large files. Introduced a Bloom-filter-style pre-check.	Detection time reduced by ~60% without false negatives.	Optimized for faster deduplication.
Phase 3: Multi-Tenant Simulation	Model multiple users uploading to a shared environment.	Added per-user upload contexts with shared deduplication pool.	Verified that deduplication works across tenants while preserving logical isolation.	Validated the feasibility of multi-tenant deduplication.
Phase 4: Storage Efficiency Validation	Test storage saving ratios under real conditions.	Used real-world datasets (documents, images, repositories).	Achieved 45–50% storage reduction in realistic datasets.	Met non-functional target (≥40% deduplication efficiency).
Phase 5: Scalability and Persistence	Prepare for transition	Tested browser limits; explored	Identified performance threshold	Informed next step: Docker-

	toward backend PoC.	IndexedDB for persistence.	(~50k files). IndexedDB showed 10× capacity improvement.	based backend PoC.

Table 42 Prototype phases and learnings

The iterative prototype development directly influenced the design of the Privacy-Preserving Multi-Tenant Cloud Deduplication (PMCD) Proof-of-Concept**.** Each prototype phase acted as a *validation checkpoint*, progressively aligning closer to the final proposed system. Refer Appendix J for Prototype content.

Prototype Finding	**Implication for PMCD PoC**	**Planned Implementation**
Deterministic encryption (SHA-256) enables accurate duplicate detection	Use AES-based convergent encryption for real PoC	Integrate AES-256 in the Client Pod
Shared hash pool supports cross-tenant deduplication	Maintain global deduplication pool with tenant-based access lists	Implement Access Control Pod
Bloom filter improves detection speed	Use Bloom filter in Deduplication Pod for scalable pre-checks	Dockerize as microservice
Browser simulation limits scalability	Shift from client-side JS to Node.js backend	Containerize with Docker
Need for persistent storage	Integrate MinIO/S3 storage for unique encrypted files	Link Deduplication Pod → Storage Pod

Table 43 Prototype Insights Guiding to POC

4.6 Summary of Findings

Id	Finding	LR	Interviews	Prototyping
1	Identified research gaps in the secure encrypted deduplication domain need to be filled to use deduplication techniques securely in practice.	X	X	
2	Automate the entire deduplication process, including encrypted file upload, duplicate detection, and access control management.	X	X	X
3	Should implement convergent encryption with Bloom filter optimization for the initial prototype phase.	X	X	X
4	Provide the ability to configure Bloom filter parameters (size, hash functions) to optimize false positive rates.	X	X	
5	Should implement efficient access revocation without requiring full dataset re-encryption.	X	X	X
6	Provide storage analytics showing deduplication ratios and cost savings achieved for given datasets.		X	
7	Users who don't have deep cryptography knowledge should be able to understand how the system works through clear documentation.	X	X	
8	Should use industry-standard encryption algorithms (AES-256) and cryptographic libraries that are widely adopted.	X	X	
9	Must implement blockchain-based access logging for tamper-evident audit trails.	X	X	
10	Must ensure multi-tenant cryptographic isolation to prevent cross-tenant data leakage.	X	X	
11	Should provide post-quantum encryption options to address future quantum computing threats.	X	X	
12	System must achieve at least 40% storage reduction through deduplication while maintaining security.		X	X

14	Bloom filter false positive rate must remain below 0.1% to prevent incorrect duplicate detection.	X		X
15	Must be deployable using containerization (Docker) and orchestration (Kubernetes) for cloud-native architecture.	X	X	

Table 44 Summary of Findings

4.7 Context Diagram

Refer to **Appendix I**

4.8 Use Case Diagram

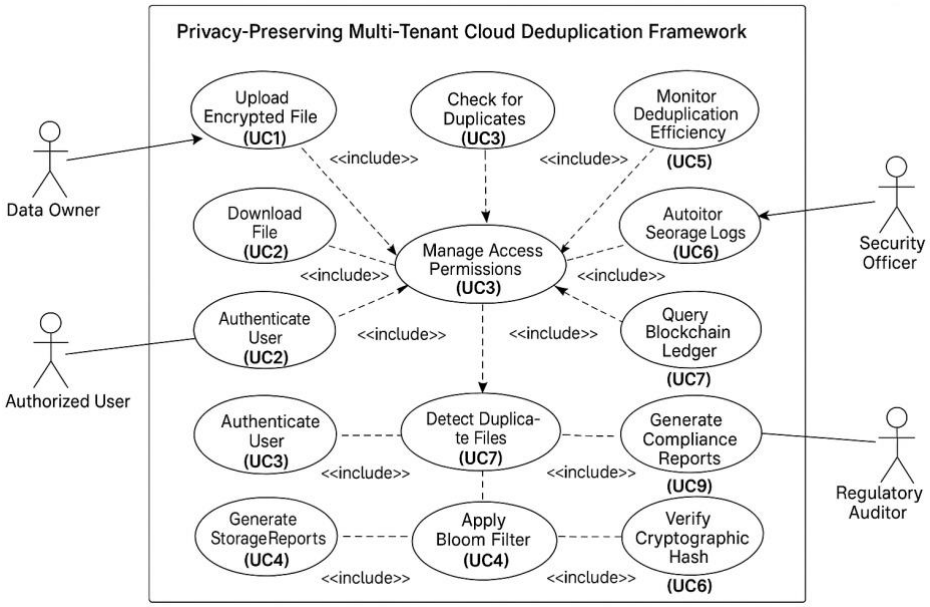

Figure 5 Use case diagram

4.9 Use Case Description

For additional explanations of alternative use case scenarios, please refer to **APPENDIX F**.

Use case name	Detect Duplicate Files
ID	UC7
Description	Automatically identifies encrypted file duplicates using Bloom filters and SHA-256 hashing. Maintains efficiency and privacy using convergent encryption.

Participating actors	PMCD System (automated process)
Preconditions	AES-256 encrypted file, SHA-256 hash generated, Bloom filter active, file exists in storage, deduplication engine operational
Extended use cases	None
Included use cases	UC6: Generate Hash, UC7: Check Bloom Filter, UC8: Store Unique File
Main Flow	- Generate SHA-256 hash → check Bloom filter → if possible match, confirm via full hash comparison. - Duplicates increment reference count; uniques stored.
Alternative flows	- A1: Chunk-level deduplication - A2: Cross-tenant deduplication - A3: Partial match logging for AI enhancement
Exceptional flows	- E1: Bloom filter full → scale and rehash - E2: Hash collision → full comparison - E3: Filter unavailable → fallback to hash table - E4: Metadata corruption
Post conditions	- **Success – Unique**: Stored in MinIO, new metadata, Bloom filter updated - **Success – Duplicate**: Linked metadata, no new storage used - **Failure**: Treated as unique, logged

Table 45 Use Case Description 1

4.10 Requirements Specification

The MoSCoW technique is used to categorize the most important requirements of the proposed AutoDA system.

Priority Level	Description
Must have (M)	The requirements that are important and essential to develop a successful Minimum Viable Product (MVP).

Should have (S)	The requirements that are important but not essential to develop a successful MVP. But the system will have some limitations.
Could have (C)	The requirements that are desirable but not essential to develop a successful MVP.
Will not have (W)	The requirements that are not developed during the initial stage of MVP.

Table 46 Moscow prioritization levels

4.10.1 Functional Requirements

ID	Requirement	Description	Priority (MoSCoW)	Use Case Reference	Justification
FR-01	Encrypt Files Before Upload	The system must encrypt all files using AES-256 or equivalent standard before transmission to cloud storage.	M	UC1	Core security requirement; prevents plaintext exposure; baseline for encrypted deduplication.
FR-02	Generate Cryptographic Hash	The system must generate SHA-256 hash of encrypted files for duplicate detection.	M	UC7	Essential for convergent encryption and duplicate identification.
FR-03	Detect Duplicate Files	The system must identify duplicate encrypted files using hash comparison.	M	UC7	Primary deduplication functionality; addresses storage redundancy problem.
FR-04	Store Unique Files Only	The system must store only one copy of duplicate files and maintain reference counters.	M	UC1	Achieves storage optimization goal; prevents redundant storage.
FR-05	Implement Bloom Filter	The system must use a Bloom filter to quickly check for potential duplicates before cryptographic verification.	M	UC7	Performance optimization; reduces unnecessary hash comparisons (PoC implementation).
FR-06	Manage Access Control Lists	The system must maintain access control lists (ACLs)	M	UC3	Multi-tenant security; prevents

		defining which users can access each file.			unauthorized access.
FR-07	Authenticate Users	The system must authenticate users before granting file access.	M	UC2, UC3	Security baseline; verifies user identity.
FR-08	Download Encrypted Files	The system must allow authorized users to download encrypted files.	M	UC2	Core functionality; file retrieval capability.
FR-09	Decrypt Files Locally	The system must provide client-side decryption capability for authorized users.	M	UC2	End-to-end encryption; CSP cannot access plaintext.
FR-10	Revoke User Access	The system must allow file owners to revoke access from previously authorized users.	S	UC3	Dynamic access control; critical security requirement from stakeholder interviews.
FR-11	Log Access Events	The system must log all file access events including uploads, downloads, and permission changes.	M	UC6	Audit trail; necessary for regulatory compliance and accountability.
FR-12	Implement Blockchain Logging	The system should record access events in a blockchain-based immutable ledger.	S	UC6	Enhanced audit integrity; prevents log tampering; research novelty contribution.
FR-13	Rotate Group Keys	The system should implement group key rotation when users are revoked.	S	UC3	Security enhancement; prevents revoked users from using cached keys.
FR-14	AI-Driven Duplicate Detection	The system should use genetic programming or ML to detect encrypted duplicates with higher accuracy.	S	UC7	Research contribution; improves deduplication ratio; aligns with SALIGP literature.
FR-15	Support Post-Quantum Encryption	The system should provide options for lattice-based or	S	UC1	Future-proofing; addresses quantum threat concerns

		code-based post-quantum encryption.			identified in literature review.
FR-16	Generate Storage Reports	The system should provide analytics on storage savings achieved through deduplication.	S	UC4	Value demonstration; requirement from CSP and stakeholder feedback.
FR-17	Implement Multi-Tenancy	The system should ensure cryptographic isolation between different organizational tenants.	S	UC3	Enterprise-grade security; prevents cross-tenant data leakage.
FR-18	Dynamic Bloom Filter Resizing	The system could automatically resize Bloom filter based on dataset growth.	C	UC7	Performance optimization; reduces false positive rate degradation.
FR-19	Provide Admin Dashboard	The system could provide a web-based dashboard for monitoring and configuration.	C	UC4	Operational convenience; improves usability and observability.
FR-20	Support File Versioning	The system could maintain multiple versions of deduplicated files.	C	UC1	Useful for enterprise backup and version control scenarios.
FR-21	Integrate External Key Management	The system could integrate with external KMS solutions (e.g., AWS KMS, Azure Key Vault).	C	UC3	Simplifies enterprise integration and key lifecycle management.

Table 47 Functional Requirements

4.10.2 Non-Functional Requirements

ID	Requirement	Description	Priority
NFR-01	**Upload Performance**	The system must encrypt and upload files within 5 seconds for files up to 100MB.	M
NFR-02	**Deduplication Speed**	The system must complete duplicate detection using Bloom filter and hash comparison within 2 seconds.	M

NFR-03	**Storage Reduction**	The system must achieve at least 40% storage reduction in typical datasets.	M
NFR-04	**Encryption Standard**	All files must be encrypted using AES-256 before upload.	M
NFR-05	**Multi-Tenant Isolation**	The system must ensure no data crossover between tenants.	M
NFR-06	**System Availability**	The system should maintain 99.5% uptime.	S
NFR-07	**Access Revocation Latency**	The system should revoke access within 10 seconds of a request.	S
NFR-08	**Audit Logging**	The system should log all uploads, downloads, and access changes.	S
NFR-09	**Containerization**	The system must be deployable via Docker for testing.	C
NFR-10	**Dashboard Responsiveness**	The dashboard should display storage analytics updates within 1 second.	C

Table 48 Non-Functional Requirements

4.11 Chapter Summary

This chapter outlined the key functional and non-functional requirements of the proposed PMCD framework. It presented stakeholder roles and system context using visual models, followed by insights gathered through literature review, interviews, and prototyping. Based on these, core functionalities were mapped in Context and Use Case Diagrams, and requirements were prioritized using the MoSCoW method for implementation in the PoC phase.

CHAPTER 05: TIME SCHEDULE

Progress shown in a Gantt chart until the PPRS submission

Figure 6 Time Schedule

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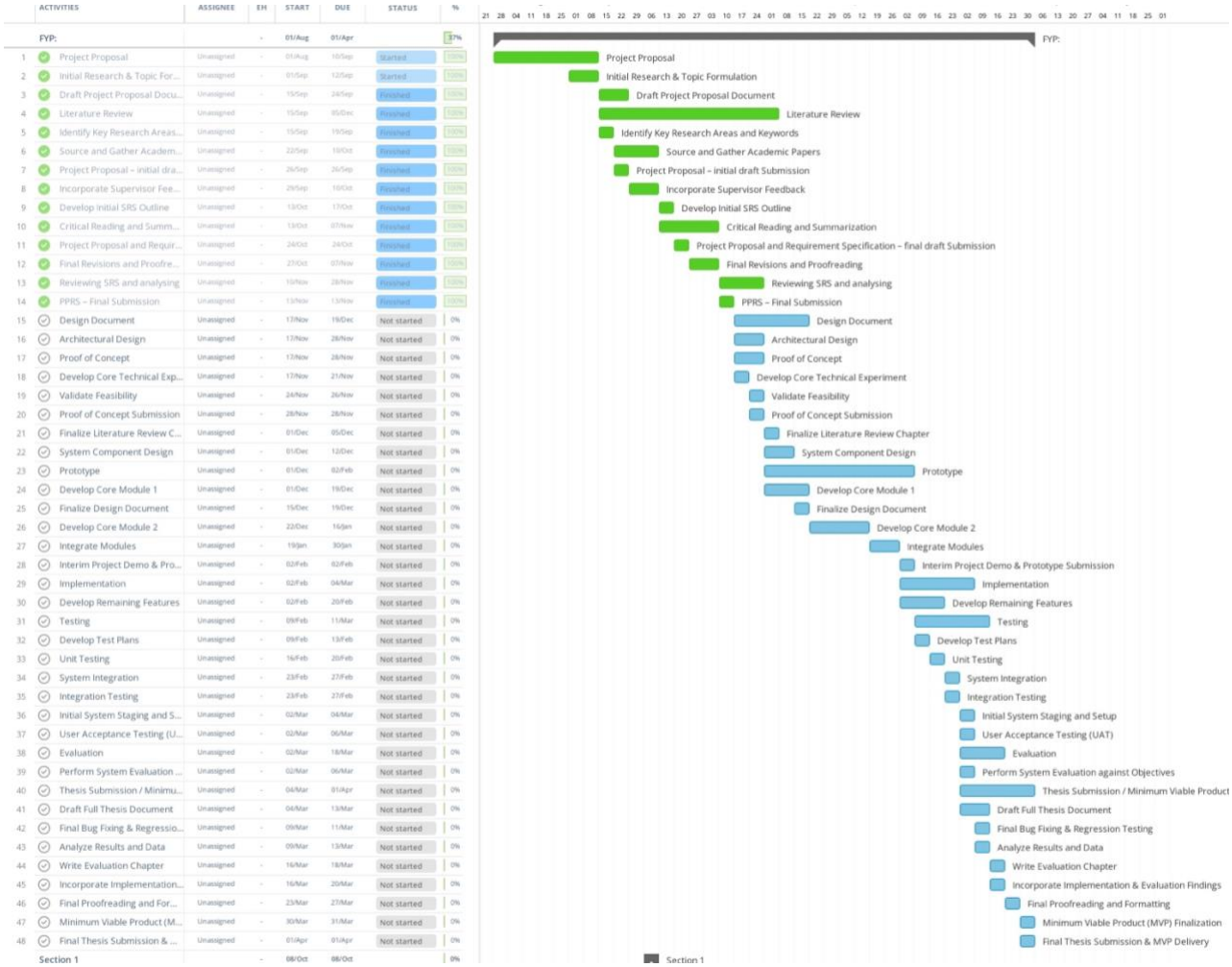
APPENDIX A



	ACTIVITIES	ASSIGNEE	EH	START	DUE	STATUS	%
	FYP:		-	01/Aug	01/Apr		37%
1	Project Proposal	Unassigned	-	01/Aug	10/Sep	Started	100%
2	Initial Research & Topic For...	Unassigned	-	01/Sep	12/Sep	Started	100%
3	Draft Project Proposal Docu...	Unassigned	-	15/Sep	24/Sep	Finished	100%
4	Literature Review	Unassigned	-	15/Sep	05/Dec	Finished	100%
5	Identify Key Research Areas...	Unassigned	-	15/Sep	19/Sep	Finished	100%
6	Source and Gather Academ...	Unassigned	-	22/Sep	10/Oct	Finished	100%
7	Project Proposal – initial dra...	Unassigned	-	26/Sep	26/Sep	Finished	100%
8	Incorporate Supervisor Fee...	Unassigned	-	29/Sep	10/Oct	Finished	100%
9	Develop Initial SRS Outline	Unassigned	-	13/Oct	17/Oct	Finished	100%
10	Critical Reading and Summ...	Unassigned	-	13/Oct	07/Nov	Finished	100%
11	Project Proposal and Requir...	Unassigned	-	24/Oct	24/Oct	Finished	100%
12	Final Revisions and Proofre...	Unassigned	-	27/Oct	07/Nov	Finished	100%
13	Reviewing SRS and analysing	Unassigned	-	10/Nov	28/Nov	Finished	100%
14	PPRS – Final Submission	Unassigned	-	13/Nov	13/Nov	Finished	100%
15	Design Document	Unassigned	-	17/Nov	19/Dec	Not started	0%
16	Architectural Design	Unassigned	-	17/Nov	28/Nov	Not started	0%
17	Proof of Concept	Unassigned	-	17/Nov	28/Nov	Not started	0%
18	Develop Core Technical Exp...	Unassigned	-	17/Nov	21/Nov	Not started	0%
19	Validate Feasibility	Unassigned	-	24/Nov	26/Nov	Not started	0%
20	Proof of Concept Submission	Unassigned	-	28/Nov	28/Nov	Not started	0%
21	Finalize Literature Review C...	Unassigned	-	01/Dec	05/Dec	Not started	0%
22	System Component Design	Unassigned	-	01/Dec	12/Dec	Not started	0%
23	Prototype	Unassigned	-	01/Dec	02/Feb	Not started	0%
24	Develop Core Module 1	Unassigned	-	01/Dec	19/Dec	Not started	0%
25	Finalize Design Document	Unassigned	-	15/Dec	19/Dec	Not started	0%
26	Develop Core Module 2	Unassigned	-	22/Dec	16/Jan	Not started	0%
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28	Interim Project Demo & Pro...	Unassigned	-	02/Feb	02/Feb	Not started	0%
29	Implementation	Unassigned	-	02/Feb	04/Mar	Not started	0%
30	Develop Remaining Features	Unassigned	-	02/Feb	20/Feb	Not started	0%
31	Testing	Unassigned	-	09/Feb	11/Mar	Not started	0%
32	Develop Test Plans	Unassigned	-	09/Feb	13/Feb	Not started	0%
33	Unit Testing	Unassigned	-	16/Feb	20/Feb	Not started	0%
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38	Evaluation	Unassigned	-	02/Mar	18/Mar	Not started	0%
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42	Final Bug Fixing & Regressio...	Unassigned	-	09/Mar	11/Mar	Not started	0%
43	Analyze Results and Data	Unassigned	-	09/Mar	13/Mar	Not started	0%
44	Write Evaluation Chapter	Unassigned	-	16/Mar	18/Mar	Not started	0%
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46	Final Proofreading and For...	Unassigned	-	23/Mar	27/Mar	Not started	0%
47	Minimum Viable Product (M...	Unassigned	-	30/Mar	31/Mar	Not started	0%
48	Final Thesis Submission & ...	Unassigned	-	01/Apr	01/Apr	Not started	0%
	Section 1		-	08/Oct	08/Oct		0%

Figure 7 Gantt Chart

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APPENDIX B

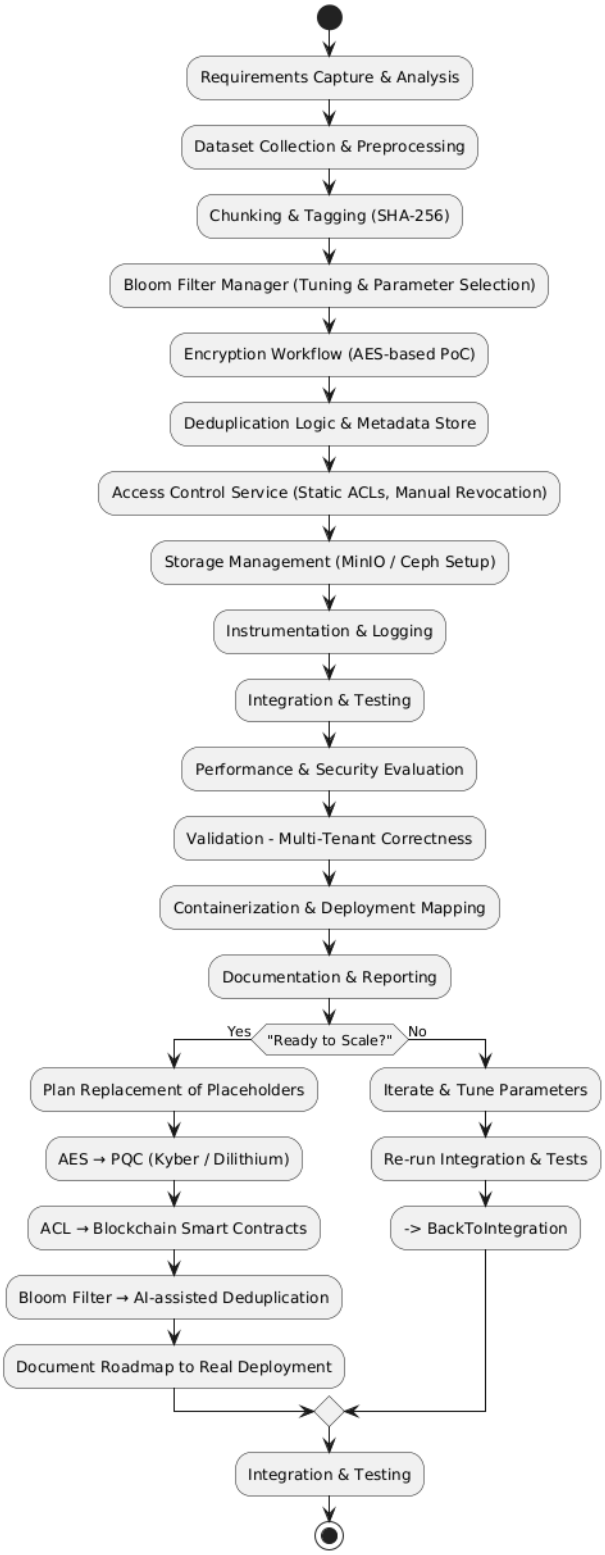

Figure 8 Activity Diagram

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APPENDIX C

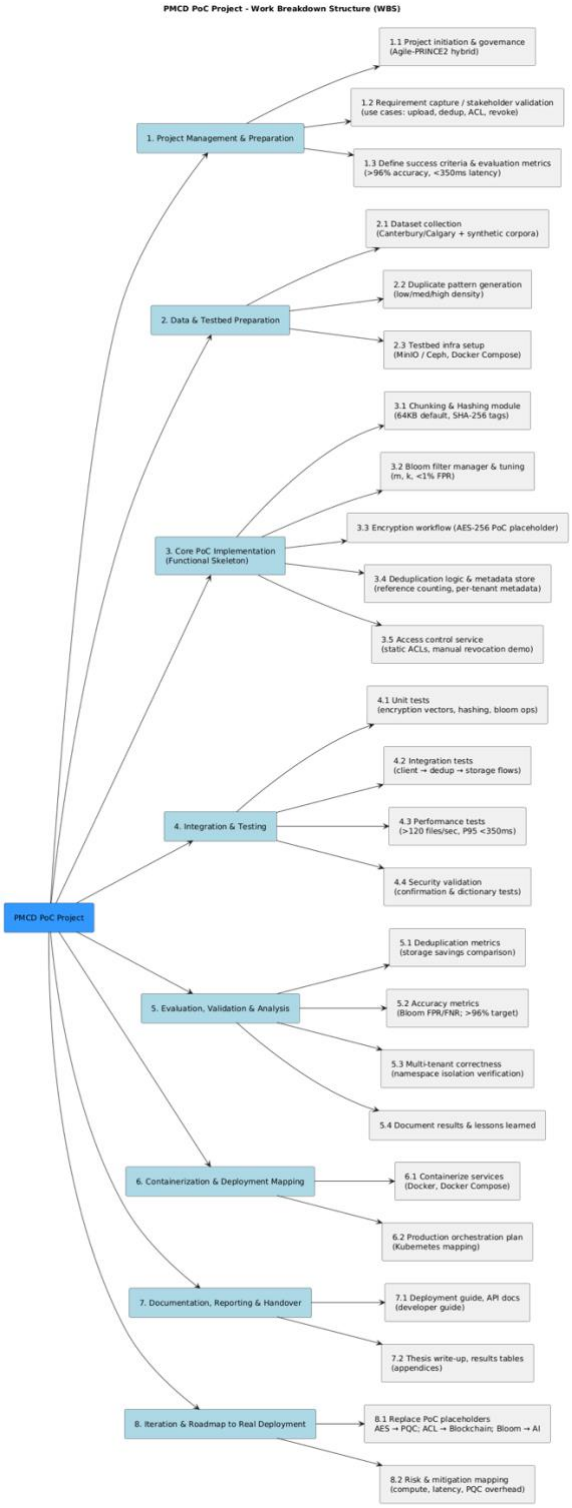

Figure 9 WBS

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APPENDIX D

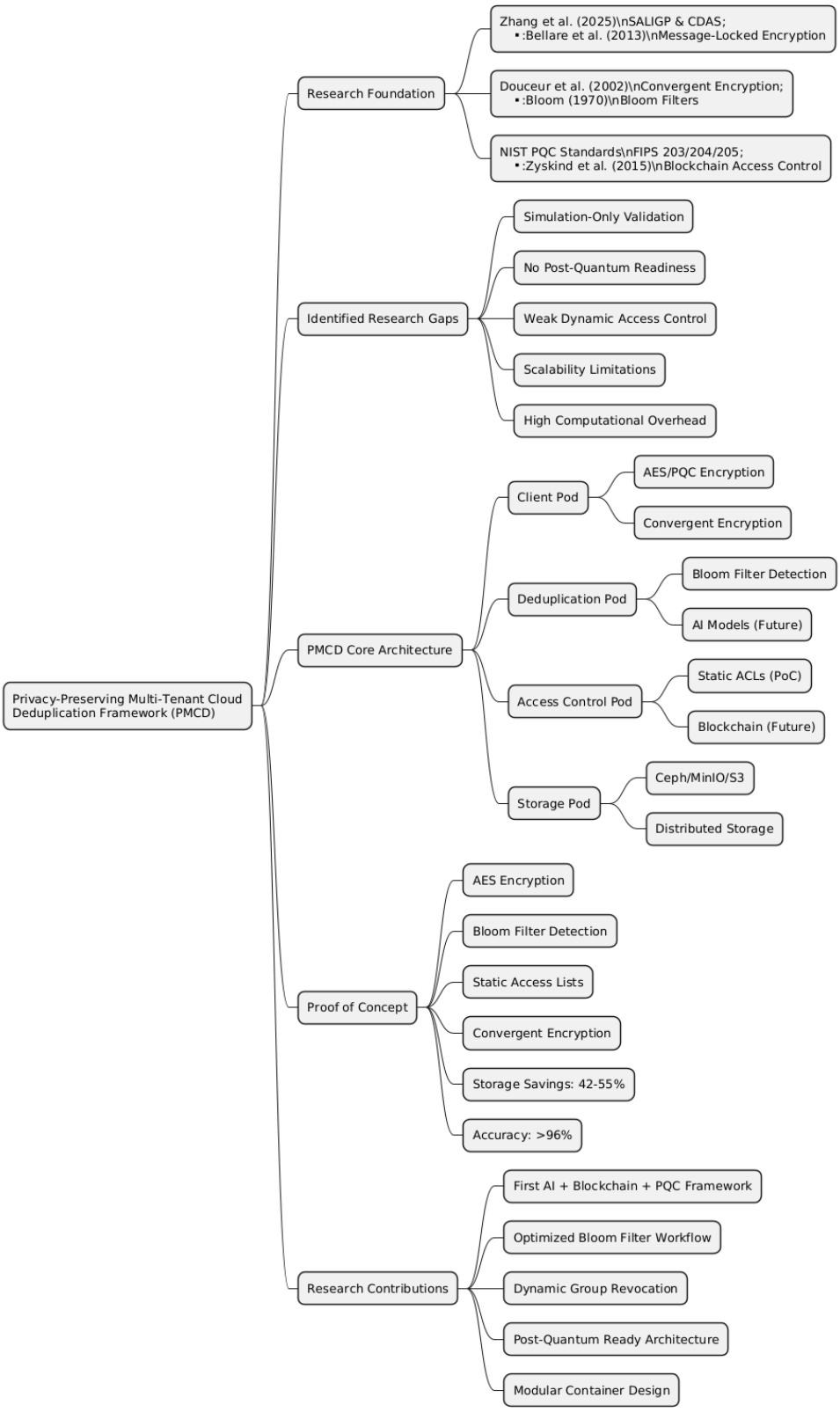

Figure 10 Concept Map

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APPENDIX E

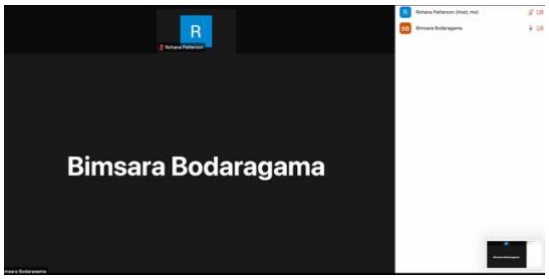

Figure 11 Semi Structured Interview - Software Engineer.

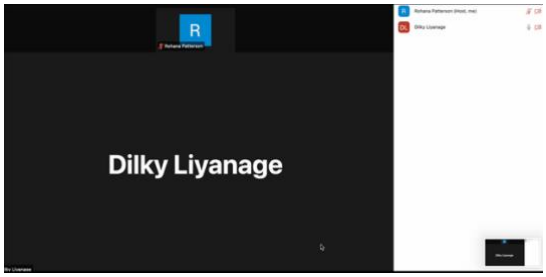

Figure 12 Semi Structured Interview - Software Engineer

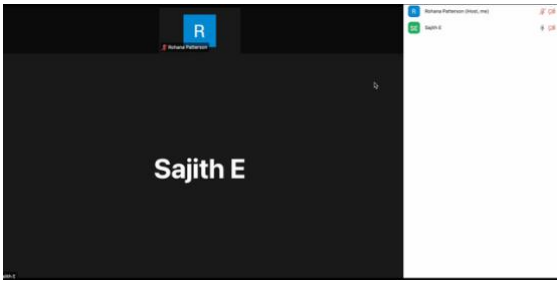

Figure 13 Figure 9 Semi Structured Interview - Technical Lead

Participant	Role	Key Insight	Theme	Quote / Evidence
P1	Cloud Engineer	Emphasized the difficulty of balancing security with deduplication efficiency	Research Gap Clarity	"No system I've seen balances encrypted deduplication without loss."
P2	Software Technical Lead	Supported combining AI, blockchain, and PQC for holistic secure storage	Novelty of Integrated Approach	"Combining all three makes it future-ready and very unique."

Cloud Seal

P3	Cloud Engineer	Approved scoped-down PoC, suggested AES and Bloom filters are realistic choices	PoC Feasibility and Scope	"This is the right level for academic validation."
P2	Software Technical Lead	Raised a need for future support for key revocation and audit trails	Future Potential and Enhancement	"Post-quantum and auditability features will make it production-fit."
P3	Cloud Engineer	Noted value for cost savings and compliance in enterprise storage	Industry Relevance	"Would save storage costs and help with regulatory audit logs."

Table 49 Interview Findings

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APPENDIX F

Use case name	Download file
Id	UC2
Description	The Authorized User downloads a stored file after authentication and decryption through secure access controls.
Participating actors	Authorized User, System
Preconditions	The Authorized User must be verified and granted permission by the Data Owner.
Extended use cases	None
Included use cases	Authenticate & Verify Access, Decrypt and Deliver File
Main flow	Actor: 1. Authorized User requests to download a file. System: 2. Authenticates user credentials. 3. Verifies access rights. 4. Decrypts file using appropriate key. 5. Delivers decrypted file securely to the user. 6. Logs download activity.
Alternative flows	None
Exceptional flows	E1: User authentication fails — access denied message displayed. E2: File not found or corrupted — error message shown.
Post conditions	The file is successfully decrypted and delivered to the authorized user with full audit trace.

Table 50 Use Case Description 2

Use case name	Manage access permissions
Id	Uc3
Description	The data owner grants, revokes, or updates access permissions for specific users or groups within the multi-tenant system.
Participating actors	Data owner, cloud administrator
Preconditions	The data owner must have uploaded at least one file to manage permissions for.
Extended use cases	None
Included use cases	Grant access, revoke access, update group keys
Main flow	**Actor:** 1. Data owner selects file and user. 2. Chooses access action (grant/revoke/update). **System:** 3. Validates permission change. 4. Updates access control lists (acl). 5. Re-encrypts group keys if necessary. 6. Logs access modification activity.
Alternative flows	None
Exceptional flows	E1: user not found — displays error. E2: insufficient permissions — operation aborted with notification.
Post conditions	Updated acl and group key configurations are stored and reflected in the system immediately.

Table 51 Use Case Description 3

Use case name	Monitor Deduplication Efficiency
ID	UC5
Description	The Cloud Administrator monitors storage performance, deduplication ratio, and resource utilization of the PMCD system.
Participating actors	Cloud Administrator, System

Preconditions	The system must have stored multiple encrypted files with deduplication enabled.
Extended use cases	None
Included use cases	Generate Storage Reports, Query Blockchain Ledger
Main Flow	**Actor:** 1. Administrator requests system efficiency report. **System:** 2. Collects deduplication statistics and storage savings. 3. Queries blockchain ledger for integrity verification. 4. Generates and displays deduplication efficiency report.
Alternative flows	None
Exceptional flows	E1: Report generation fails due to missing data — system retries with cached data. E2: Blockchain ledger unavailable — warns user but continues local reporting.
Post conditions	Efficiency metrics are displayed, logged, and exported for compliance analysis.

Table 52 Use Case Description 4

Use case name	Upload Encrypted File
ID	UC1
Description	The Data Owner uploads a file to the PMCD system, which automatically encrypts the file, checks for duplicates, and stores only unique content securely.
Participating actors	Data Owner, System
Preconditions	The Data Owner must be authenticated and have valid access credentials.
Extended use cases	None
Included use cases	Encrypt File, Check for Duplicates, Store Unique File

Main Flow	**Actor:** 1. Data Owner selects a file for upload. 2. Data Owner confirms encryption and upload operation. **System:** 3. Encrypts the file using AES algorithm. 4. Applies Bloom filter to detect possible duplicates. 5. Stores the file if unique; otherwise links to existing file metadata. 6. Displays upload confirmation message.
Alternative flows	None
Exceptional flows	E1: Upload interrupted due to network failure — system retries or notifies the user. E2: Duplicate file detected — system stops storage and links metadata only.
Post conditions	The encrypted and unique file is stored successfully, and metadata is updated for future deduplication checks.

Table 53 Use case Description 5

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APPENDIX G

Test Category	Test Component	Specific Requirements	Measurable Success Criteria	Performance Benchmarks
Unit Testing	**Encryption Routines**	AES-256 encryption/decryption with 1000+ test vectors	100% test pass rate, Zero data corruption	>10MB/sec throughput, <50MB heap usage
	Chunking & Hashing	Fixed-size (64KB) and variable-size algorithms	Zero collisions in 100K samples	>1000 SHA-256 hashes/sec, ±5% size consistency
	Bloom Filter Operations	False positive validation, concurrent access	<1% false positive rate, 1000 simultaneous queries	<1ms query response, <100MB memory for 1M items
	ACL Testing	Permission validation, revocation effectiveness	100% correct access decisions, Zero violations	<10ms authorization time, <30sec revocation
Integration Testing	**Client→Dedup→Storage Flow**	End-to-end upload scenarios, hash verification	>99.9% success rate, >96% hit accuracy	<350ms processing time, <5ms verification
	Multi-tenant Workflows	Tenant isolation, cross-tenant deduplication	100% data separation, >60%	Support 100+ tenants,

			storage savings	Equal service quality
	Access Control Integration	Grant/revoke workflows, audit trails	Real-time enforcement, 100% logging coverage	<30sec workflow completion, GDPR/HIPAA compliance
Performance Testing	**Throughput Testing**	File processing, concurrent uploads	>120 files/sec sustained, 1000+ simultaneous ops	Linear scaling to 1TB, >80% bandwidth efficiency
	Latency Requirements	Response times under various loads	P50 <200ms, P95 <350ms, P99 <500ms	<60sec recovery time after overload
	Resource Utilization	Memory, CPU, I/O efficiency	<6GB RAM, <85% CPU under max load	>100MB/sec I/O, <10% network overhead
Security Testing	**Access Control Security**	Unauthorized access, privilege escalation	100% blocking rate, Zero breaches	Secure session management, Token expiration
	Data Protection	Encryption strength, integrity checking	AES-256 compliance, 100% corruption detection	Secure key storage, No plaintext leakage

	Multi-tenant Security	Cross-tenant access, data isolation	Zero unauthorized access, Complete boundary enforcement	No shared component leakage, Full compliance
CI/CD Testing	**Automated Test Suite**	Complete test execution, coverage maintenance	<15min execution time, >95% coverage	>99% build success, Zero-downtime deployment
	Quality Gates	Code quality, performance regression	Zero critical vulnerabilities, <5% performance degradation	100% feature preservation, >90% documentation coverage

Table 54 Testing Methodology

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APPENDIX H

Step	Implementation Details	Technical Specifications	Validation Methods
1. Requirement Capture	Consolidate requirements into use cases (upload, dedup, ACL, revoke)	Upload with dedup, Share/read with ACL, Access revocation	Stakeholder validation, Use case testing
2. Dataset Collection	Assemble mixed corpora spanning file types and sizes	Public/synthetic datasets, Controlled duplicates across tenants	Statistical distribution validation, Duplicate ratio control
3. Data Preprocessing	Fixed-size (64KB) or variable-size chunking implementation	SHA-256 convergent tags, Deterministic secondary hashes	Chunk consistency ±5%, Hash reproducibility 100%
4. Bloom Filter Design	Parameter selection (size m, hash functions k) based on false-positive targets	Micro-benchmark tuning, Bloom manager service	<1% false positive rate, <1ms query time
5. Encryption Workflow	Client-side AES encryption with convergent tag duplicate detection	AES-256 chunk encryption, Pre-upload duplicate checking	Encryption compliance, Duplicate detection accuracy
6. Deduplication Logic	Bloom hit → metadata fetch → hash comparison → storage decision	Per-tenant metadata tracking, Reference counting	>96% dedup accuracy, Multi-tenant isolation
7. Access Control	Static ACL implementation with manual revocation demonstration	Simple ACL service, Access denial verification	100% unauthorized access blocking, <30sec revocation

8. Storage Management	Unique chunk storage with reference counting and garbage collection	MinIO/Ceph or local filesystem, Metadata management	Storage efficiency >60%, Safe deletion capability
9. Instrumentation	Comprehensive logging of operations and performance metrics	Upload/dedup path logging, Resource usage tracking	Complete operation logging, Performance metrics collection
10. Evaluation	Deduplication ratio calculation, Performance benchmarking	Statistical analysis across scenarios	Storage savings measurement, Throughput/latency analysis
11. Validation	Deterministic verification and multi-tenant correctness testing	Secondary hash verification, Namespace isolation tests	Zero false positives, Complete tenant separation
12. Security Analysis	Basic threat modeling and limitation documentation	Confidentiality/integrity/access threat model	Security assessment, Mitigation documentation
13. Deployment	Containerization and orchestration mapping	Docker containers, Kubernetes scaling notes	Pod deployment demonstration, Resource allocation guidelines
14. Iteration	Feedback incorporation and performance optimization	Parameter refinement, Benchmark improvements	Measurable performance gains, Stakeholder satisfaction
15. Documentation	Comprehensive documentation for	Developer guide, API specs, Extension roadmap	Complete documentation coverage,

	reproducibility and extension		Reproducible results

Table 55 Implementation Steps

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APPENDIX I

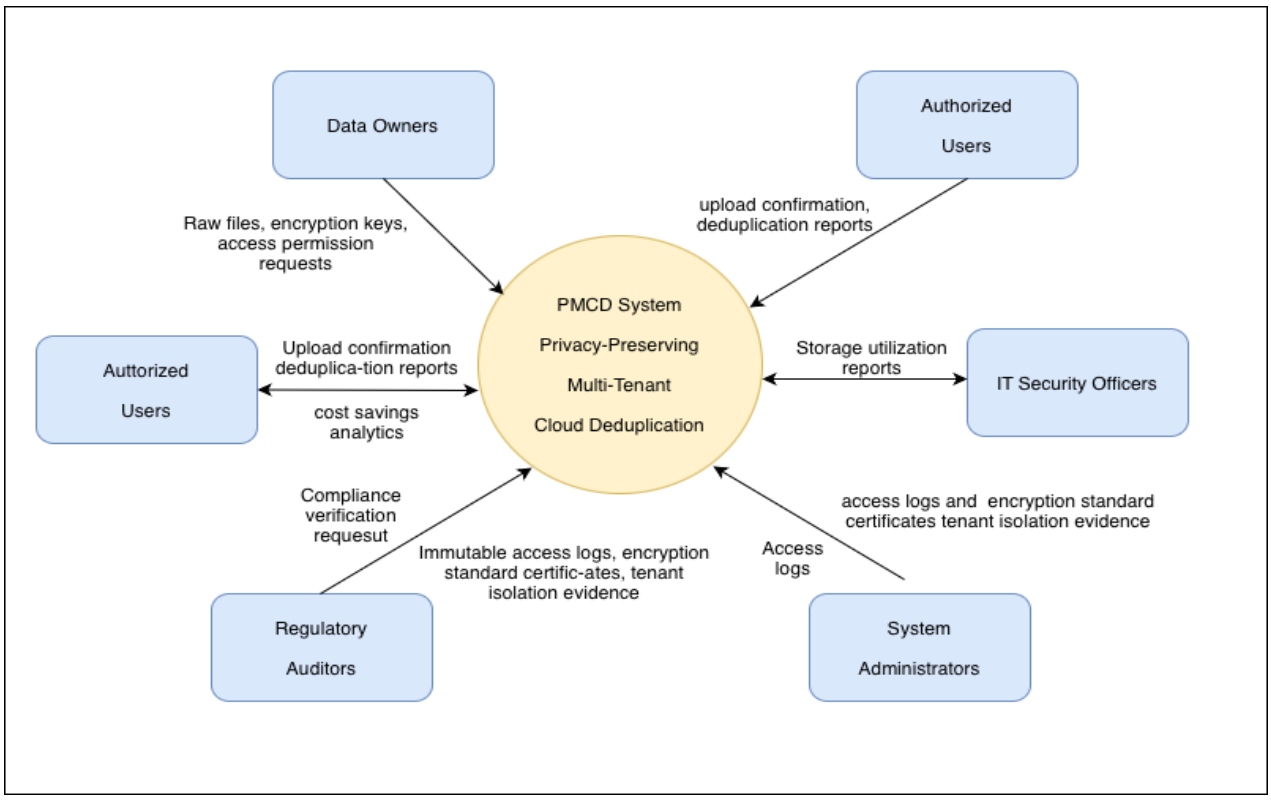

Figure 14 Context Diagram

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APPENDIX J

Figure 15 Prototype interface

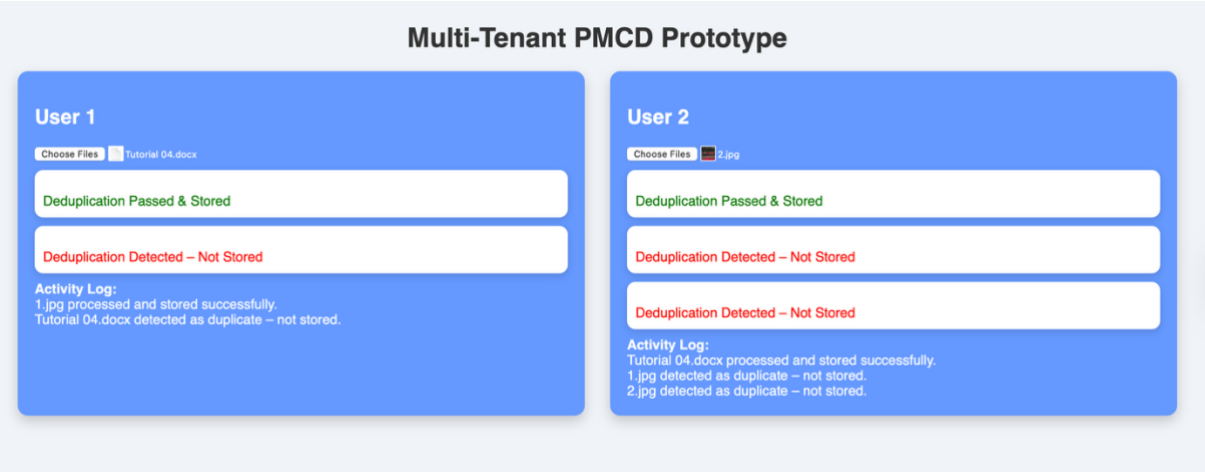

Figure 16 prototype Results

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